

# MADRE

Agentic  
System

• MODEL-AGNOSTIC • DELAYED • REASONING • EFFORT • [Beta v0.1]

---

## Project Definition and Architecture Documentation

---

Scvria Software  
June 6, 2026

# Contents

|                                                    |           |
|----------------------------------------------------|-----------|
| <b>Front Matter</b>                                | <b>1</b>  |
| 1.1 Executive Definition . . . . .                 | 1         |
| <b>Research Foundation</b>                         | <b>5</b>  |
| 2.1 Research Problem . . . . .                     | 5         |
| 2.2 Research Questions . . . . .                   | 5         |
| 2.3 Hypotheses . . . . .                           | 6         |
| 2.4 Expected Contributions . . . . .               | 6         |
| 2.5 Novelty And Validation Strategy . . . . .      | 6         |
| 2.6 Related Work And Evidence Map . . . . .        | 7         |
| <b>Product Definition</b>                          | <b>8</b>  |
| 3.1 Executive Definition . . . . .                 | 8         |
| 3.2 Product Rationale . . . . .                    | 8         |
| 3.2.1 Extended Rationale . . . . .                 | 8         |
| 3.3 Product Design . . . . .                       | 10        |
| 3.3.1 Product Scope . . . . .                      | 10        |
| 3.3.2 What <b>MADRE</b> Is . . . . .               | 10        |
| 3.3.3 What <b>MADRE</b> Is Not . . . . .           | 10        |
| 3.3.4 Core User Value . . . . .                    | 11        |
| 3.3.5 Design Principles . . . . .                  | 11        |
| 3.3.6 Product Boundaries . . . . .                 | 11        |
| <b>Operational Scenarios</b>                       | <b>13</b> |
| 4.1 Scenario Cards . . . . .                       | 13        |
| <b>Conceptual Architecture</b>                     | <b>14</b> |
| 5.1 Formal Requirements . . . . .                  | 14        |
| 5.2 High-Level Architecture . . . . .              | 14        |
| 5.2.1 Conceptual Components . . . . .              | 16        |
| 5.2.2 Responsibility Layer Diagram . . . . .       | 16        |
| 5.2.3 Concurrent Agent Lanes . . . . .             | 17        |
| 5.3 Runtime Data Flow . . . . .                    | 17        |
| 5.3.1 Canonical Request Narrative . . . . .        | 18        |
| 5.3.2 Governed Model And Action Sequence . . . . . | 18        |
| 5.3.3 Data And Authority Rules . . . . .           | 21        |
| 5.3.4 Trust Boundary Diagram . . . . .             | 22        |
| 5.3.5 Access Surfaces . . . . .                    | 22        |

|                                                |           |
|------------------------------------------------|-----------|
| <b>Safety And Evolution Policy</b>             | <b>24</b> |
| 6.1 Runtime Safety . . . . .                   | 24        |
| 6.1.1 Policy Before Action . . . . .           | 24        |
| 6.1.2 Autonomy Levels . . . . .                | 24        |
| 6.2 Evolution Control . . . . .                | 25        |
| 6.2.1 Evolution Rules . . . . .                | 25        |
| <b>System Engineering Baseline</b>             | <b>26</b> |
| 7.1 Engineering Problem . . . . .              | 26        |
| 7.2 Stakeholders And Concerns . . . . .        | 26        |
| 7.3 Requirement Domains . . . . .              | 26        |
| 7.4 Traceability Surface . . . . .             | 27        |
| 7.5 Quality Attributes . . . . .               | 27        |
| 7.6 Risk And Policy Boundaries . . . . .       | 27        |
| <b>Requirements Baseline</b>                   | <b>28</b> |
| 8.1 Requirement Taxonomy . . . . .             | 28        |
| 8.2 Functional Requirements . . . . .          | 29        |
| 8.3 Cross-Cutting Requirements . . . . .       | 30        |
| <b>Traceability Model</b>                      | <b>33</b> |
| 9.1 Traceability Method . . . . .              | 33        |
| 9.2 Architecture Traceability Matrix . . . . . | 33        |
| 9.3 Integrated Architecture Trace . . . . .    | 34        |
| <b>Design Decision Traceability</b>            | <b>36</b> |
| <b>Architecture Viewpoints</b>                 | <b>38</b> |
| 11.1 Viewpoint Catalogue . . . . .             | 38        |
| 11.2 Viewpoint Design Focus . . . . .          | 39        |
| <b>Runtime Governance</b>                      | <b>41</b> |
| 12.1 Runtime Authority Rule . . . . .          | 41        |
| 12.2 Policy And Action Gate . . . . .          | 41        |
| 12.3 Policy Matrix . . . . .                   | 41        |
| 12.4 Learning Lifecycle . . . . .              | 42        |
| <b>Threat Model</b>                            | <b>43</b> |
| 13.1 Threat Taxonomy . . . . .                 | 43        |
| <b>Quality Model</b>                           | <b>45</b> |
| 14.1 Quality Attribute Scenarios . . . . .     | 45        |
| <b>UX, Observability, And Recovery</b>         | <b>47</b> |
| 15.1 Interaction Contract . . . . .            | 47        |
| 15.2 Observability Surfaces . . . . .          | 47        |
| <b>Evaluation And Benchmark Specification</b>  | <b>48</b> |
| 16.1 Benchmark Families . . . . .              | 48        |

|                                                           |           |
|-----------------------------------------------------------|-----------|
| <b>Governance, Deployment, And Limits</b>                 | <b>50</b> |
| 17.1 Secure Engineering Principles . . . . .              | 50        |
| 17.2 Open-Source Governance . . . . .                     | 50        |
| 17.3 Deployment And Operational Assumptions . . . . .     | 50        |
| 17.4 Risks, Limitations, And Excluded Functions . . . . . | 50        |
| <b>Requirements Catalogue</b>                             | <b>51</b> |
| <b>Traceability Matrix</b>                                | <b>52</b> |
| <b>Threat Catalogue</b>                                   | <b>53</b> |
| <b>Benchmark Catalogue</b>                                | <b>54</b> |
| <b>Runtime Domain Model</b>                               | <b>55</b> |
| 22.1 Design Rule . . . . .                                | 55        |
| 22.2 Minimal Canonical Flow . . . . .                     | 55        |
| 22.3 Conventions . . . . .                                | 56        |
| 22.3.1 Type Conventions . . . . .                         | 56        |
| 22.3.2 Process Conventions . . . . .                      | 56        |
| 22.3.3 Class Inventory . . . . .                          | 56        |
| 22.3.4 Structural Relations . . . . .                     | 57        |
| 22.4 Class Definitions . . . . .                          | 58        |
| 22.4.1 MADREKernel . . . . .                              | 58        |
| 22.4.2 ReasoningModule . . . . .                          | 59        |
| 22.4.3 MADREAgent . . . . .                               | 60        |
| 22.4.4 SimpleAgent . . . . .                              | 61        |
| 22.4.5 TwinAgent . . . . .                                | 62        |
| 22.4.6 ComplexAgent . . . . .                             | 62        |
| 22.4.7 SystemAgent . . . . .                              | 63        |
| 22.4.8 ReasoningRequest . . . . .                         | 64        |
| 22.4.9 ReasoningResponse . . . . .                        | 65        |
| 22.4.10 ReasoningPlan . . . . .                           | 65        |
| 22.4.11 ReasoningTask . . . . .                           | 66        |
| 22.4.12 AgentRequest . . . . .                            | 67        |
| 22.4.13 AgentResponse . . . . .                           | 68        |
| 22.4.14 MADREAction . . . . .                             | 69        |
| 22.4.15 Workflow . . . . .                                | 70        |
| 22.4.16 ContextBundle . . . . .                           | 70        |
| 22.4.17 ReasoningArtifact . . . . .                       | 71        |
| 22.4.18 KnowledgeRecord . . . . .                         | 72        |
| 22.4.19 BoundaryContract . . . . .                        | 73        |
| 22.4.20 PolicyDecision . . . . .                          | 74        |
| 22.4.21 RuntimeJournal . . . . .                          | 74        |
| 22.4.22 DBAccessor . . . . .                              | 75        |
| 22.4.23 RuntimeEvent . . . . .                            | 76        |
| 22.5 Validation Checklist . . . . .                       | 76        |
| <b>Glossary</b>                                           | <b>78</b> |

# Front Matter

## Abstract

### 1.1 Executive Definition

*"MADRE is a local-first, model-agnostic runtime kernel for building agentic systems that preserve user sovereignty while coordinating reasoning modules, module-owned agents, governed context, bounded actions, known material, evaluated learning, recovery, and inspectable execution."*

"

**MADRE** Agentic System[Beta v0.1] exists because the current AI market is normalizing a dangerous bargain: users receive convenient language interfaces, while their data, dependency chain, costs, and decision power move into remote platforms they do not control. That bargain is technically unnecessary and politically unacceptable for many individuals, teams, public-interest organizations, researchers, creators, and small developers.

The project does not treat a language model as a complete intelligence. A model is a powerful linguistic function: useful, fallible, statistically trained, expensive in context, and unsafe when it is allowed to become the whole system. **MADRE** puts that function inside an engineered runtime. The runtime decides what may be known, what may be sent, what may be stored, what may be acted on, and what must wait for stronger evidence or human authority.

The product objective is one coherent assistant experience backed by a durable kernel. Visible interaction can remain fast and conversational while module-local reasoning and reflection lanes decompose, schedule, check, record, and resume deeper work. This is the practical meaning of Model Agnostic Delayed Reasoning Effort: useful local intelligence should come from architecture, context discipline, and bounded autonomy, not from blind dependence on one remote provider or one large model.

**MADRE** is therefore not a chatbot, not a prompt collection, not a wrapper around one vendor API, and not a task-management application. It is the runtime layer that applications use to coordinate models, internal actions, conversation and knowledge records, policies, and interfaces without surrendering the user's private domain to cloud-first AI infrastructure.

## Specification Scope

**MADRE** is a local-first, model-agnostic runtime kernel for constructing agentic systems whose useful behavior is produced by software architecture rather than by treating language models as autonomous intelligence. The runtime assigns a Core Module, permits its **MainAgent** to provide foreground continuity, governs context before **ModelAction** or other actions, keeps **KnowledgeRecord** handling inspectable,

records local traces through `RuntimeJournal`, and requires learning to pass through evaluation, promotion, and rollback boundaries.

This specification defines the research, product, and architecture baseline for MADRE. It states the system problem, expected contribution, stakeholder concerns, requirement domains, architecture views, safety model, policy model, and evaluation surface used to assess runtime behavior. The product is an open, extensible foundation for private and affordable agentic software on ordinary local devices, with optional remote services and integrations treated as governed risk boundaries rather than defaults.

The scope of MADRE comprises runtime purpose, system boundary, requirement families, architecture obligations, risk model, quality attributes, validation strategy, and product-level authority semantics.

The design identifiers D-001 through D-013 are internal traceability anchors inside this dossier. They identify product rationale, runtime obligations, safety boundaries, and evaluation surfaces at architecture level without making prior side specifications authoritative.

## Canonical Terms

| Term                   | Meaning in MADRE                                                                                                                                                                                                                                                             |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MADREKernel            | Deterministic local coordinator that registers <code>ReasoningModule</code> objects, assigns the Core Module, routes requests, schedules work, coordinates resources, applies policy, and records runtime activity through <code>RuntimeJournal</code> .                     |
| ReasoningModule        | Governed reasoning unit for a bounded domain or scope. It owns module records, workflows, routines, actions, agents, knowledge and learning boundaries, and allowed model/inference scope.                                                                                   |
| Core Module            | Kernel registration role held by a compatible <code>ReasoningModule</code> that receives ordinary interaction first; it is not a separate module type.                                                                                                                       |
| MADREAgent / MainAgent | A <code>MADREAgent</code> is a concrete module-owned runtime actor constrained by <code>ModelBinding</code> ; <code>MainAgent</code> is the role held by one module-owned <code>ComplexAgent</code> , while other <code>ComplexAgent</code> objects need not hold that role. |

| Term                                                                       | Meaning in MADRE                                                                                                                                                                                                   |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SimpleAgent, TwinAgent, ComplexAgent                                       | Agent profiles for direct execution, review/evaluation over a produced result, and foreground/reasoning/reflection capability respectively; none gains authority outside its owning module.                        |
| ReasoningPlan, Workflow, AgentRoutine                                      | A request-derived reasoning plan may select reusable module workflow and agent routine capabilities without turning either into a second plan or a new authority layer.                                            |
| Access surface                                                             | A CLI, local console, GUI, speech surface, or application interface that exposes the kernel without becoming runtime authority.                                                                                    |
| ReasoningTask, RuntimeEvent, RuntimeJournal                                | <b>ReasoningTask</b> tracks delayed-work lifecycle; <b>RuntimeEvent</b> is append-only runtime history; <b>RuntimeJournal</b> is the local trace mechanism that stores or references them and related payloads.    |
| PolicyDecision                                                             | Explicit allow, block, or authorization-required outcome for an affected action or data class and authority boundary.                                                                                              |
| Foreground lane                                                            | A possible <b>ComplexAgent</b> capability used by a module's <b>MainAgent</b> to preserve flow, clarify intent, answer within safe context, and create delayed work when stronger evidence is required.            |
| Reasoning lane                                                             | Durable module-local work used for analysis, evidence gathering, verification, internal action invocation, or higher-effort inference.                                                                             |
| ContextSource / ContextBundle                                              | Source descriptor and bounded, classified, minimized, revocable selection of material prepared for an agent request, action, or inference execution.                                                               |
| AgentAction, ModelAction, ModelBinding, InferenceProfile, InferenceBackend | An executable capability and its inference-backed specialization; a model-backed action executes only through the binding allowed for its module-owned agent, with configured and replaceable inference execution. |

| Term                                   | Meaning in MADRE                                                                                                                                                                                                        |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| InferenceRecord / InferenceOutput      | One recorded inference execution and its generated material. <b>InferenceOutput</b> is not truth, policy, memory, knowledge, behavior, routing, authority, or a candidate by itself.                                    |
| KnowledgeRecord / KnowledgeCandidate   | Known material held by a module with provenance, scope, sensitivity, intended use, truth-level/authority, correction, and lifecycle metadata; only a pending knowledge-boundary change is a <b>KnowledgeCandidate</b> . |
| LearningCandidate / ConversationRecord | An evaluated proposed module improvement and retained interaction material respectively. Saving known material is not learning, and conversation is not knowledge by default.                                           |
| Artifact                               | A materialized output such as a file, document, code patch, report, image, dataset, PDF, snapshot, or final buffer/document. Ordinary conversation text is not an artifact unless materialized.                         |

## Specification Baseline

| Version   | Status                                                                                                                                                                                                                                      |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Beta v0.1 | Research, product, and architecture specification baseline for local-first, model-agnostic delayed reasoning. The baseline covers product requirements, architecture obligations, safety boundaries, traceability, and evaluation criteria. |



# Research Foundation

---

## 2.1 Research Problem

Modern language-model systems often combine private data, model calls, action execution, retained interaction, known material, and remote infrastructure inside opaque conversational interfaces. This makes useful demonstrations easy to produce, but it weakens local ownership, traceability, autonomy control, error recovery, and the ability to replace models or providers.

**MADRE** addresses this as a Software Engineering problem: how to design a local-first runtime in which language models remain fallible replaceable functions, while the software system owns policy, context, knowledge, internal actions, runtime trace, recovery, and learning boundaries. The research problem is not to prove that models reason, but to define an architecture that extracts useful language behavior without surrendering system authority to **InferenceOutput**.

## 2.2 Research Questions

| ID   | Research question                                                                                                                                                                                                                                                    |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RQ-1 | How can a local-first runtime separate immediate interaction from delayed reasoning without losing continuity, accountability, or user control?                                                                                                                      |
| RQ-2 | How can model-agnostic contracts prevent provider-specific assumptions from entering policy, knowledge handling, action authority, and truth management?                                                                                                             |
| RQ-3 | What traceability structure is sufficient to connect <b>MADRE</b> 's product claims, stakeholder concerns, runtime requirements, architecture decisions, and validation evidence?                                                                                    |
| RQ-4 | How can context, <b>ConversationRecord</b> , <b>KnowledgeRecord</b> , <b>KnowledgeCandidate</b> , and <b>LearningCandidate</b> remain useful while preserving provenance, truth authority, minimization, revocation, correction, promotion, and rollback boundaries? |
| RQ-5 | What integrated behavior demonstrates that <b>MADRE</b> is a governed runtime rather than a conversational wrapper?                                                                                                                                                  |

## 2.3 Hypotheses

| ID  | Hypothesis                                                                                                                                                                                                                                       |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| H-1 | Module-local foreground, reasoning, and reflection lanes can preserve conversational responsiveness while improving reliability for tasks that require evidence, verification, action execution, or stronger context.                            |
| H-2 | Model-agnostic contracts can reduce provider lock-in when policy, knowledge handling, context, and action execution remain owned by the runtime.                                                                                                 |
| H-3 | Context governance with provenance, classification, minimization, and revocation can reduce leakage and misuse without eliminating useful specialized reasoning.                                                                                 |
| H-4 | Evaluated <code>LearningCandidate</code> promotion can improve system adaptation while preventing <code>InferenceOutput</code> or ordinary <code>KnowledgeRecord</code> storage from silently rewriting protected behavior or factual authority. |
| H-5 | <code>RuntimeJournal</code> , <code>ReasoningTask</code> , and append-only <code>RuntimeEvent</code> history can make delayed reasoning inspectable and recoverable enough for ordinary local operation.                                         |

## 2.4 Expected Contributions

**MADRE**’s expected contribution is a reference architecture for local-first, model-agnostic delayed reasoning. The contribution combines module-owned agents, optional Complex Agent lane profiles, governed context bundles, explicit `ModelAction` and internal action contracts, `InferenceRecord`/`InferenceOutput` separation, controlled learning, and traceable recovery paths.

The novelty is not a new language model. It is the engineering arrangement that keeps language models useful while preventing them from becoming sources of truth, authority, policy, knowledge, or action. The architecture is intended to be useful for personal assistants, developer systems, organizational knowledge workers, domain systems, and local interfaces that require control over private knowledge.

## 2.5 Novelty And Validation Strategy

The first research claim is that a useful agentic system can be validated as a software architecture before it is validated as a broad automation product. **MADRE**’s novelty is the conjunction of local-first execution, replaceable inference and action boundaries, delayed reasoning, governed context, explicit `PolicyDecision` outcomes, evaluated learning promotion, and journaled recovery.

The validation strategy examines whether a local request can enter through bounded access surfaces, be triaged without unsupported claims, become a durable `ReasoningTask` when stronger evidence is needed, obtain a governed context bundle, cross a replaceable `InferenceBackend`, invoke a bounded internal action after a permitting `PolicyDecision`, produce an `InferenceOutput` that remains non-authoritative, expose `RuntimeJournal` trace evidence, and support correction, invalidation, supersession, rollback, or deletion/detachment.

The validation strategy treats negative paths as first-class evidence. Prompt or source injection must remain source data. Known material used with an incorrect `truthLevel` or `truthAuthority` must be rejected or corrected. Sensitive or unrelated material must be excluded from context. `InferenceOutput`

must not become truth, policy, action authority, memory, knowledge, behavior, routing, or a candidate by itself. A **KnowledgeCandidate** changes a knowledge boundary only through promotion, and a **LearningCandidate** changes module behavior only after defined evaluation and promotion.

## 2.6 Related Work And Evidence Map

| Evidence family          | Relevance to MADRE                                                                                                                                                 | Specification use                                              |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Local-first software     | Establishes user ownership, offline usefulness, privacy, longevity, and optional synchronization as product concerns.                                              | Supports data sovereignty and local-first architecture claims. |
| Requirements engineering | Provides structure for measurable requirements, acceptance criteria, verification methods, and source traceability.                                                | Supports the SRS and traceability model.                       |
| Architecture description | Provides stakeholder, concern, viewpoint, view, correspondence, and decision-rationale concepts.                                                                   | Supports multi-view architecture description.                  |
| LLM and agent security   | Identifies prompt injection, sensitive disclosure, supply-chain risk, data poisoning, excessive agency, vector weakness, misinformation, and resource consumption. | Supports threat and policy modeling.                           |
| Software quality models  | Provide vocabulary for reliability, maintainability, portability, security, performance efficiency, usability, and quality-in-use evaluation.                      | Supports the quality attribute model.                          |
| AI evaluation research   | Provides multi-metric and evidence-oriented evaluation patterns for generated material, retrieval quality, and agentic task performance.                           | Supports MADRE-native benchmark families.                      |

# Product Definition

## 3.1 Executive Definition

**MADRE** is a runtime foundation rather than a single assistant interface. The kernel assigns ordinary interaction to a Core Module, records delayed lifecycle and runtime history locally, and keeps policy, knowledge, action, and recovery authority outside inference. Language models run through module-constrained **ModelAction** execution; each **InferenceOutput** is generated material from an **InferenceRecord**, not authority or a candidate by itself.

## 3.2 Product Rationale

### MADRE Agentic System

| Rationale                   | Product implication                                                                                                                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Local control               | Private data, local knowledge, and sensitive work remain under local authority unless a policy-governed transfer is explicitly permitted.                                                                              |
| Model substitutability      | No model provider, host, prompt convention, or output format defines runtime truth, policy, knowledge, or action authority.                                                                                            |
| Delayed reasoning           | Work that requires evidence, verification, resource control, or stronger context is represented as a traceable <b>ReasoningTask</b> rather than as an unsupported foreground answer.                                   |
| Context governance          | Context is selected, minimized, classified, sourced, scoped, provenanced, and revocable before it is provided to a <b>ModelAction</b> , internal action, or reasoning task.                                            |
| Bounded action use          | Internal actions, external actions, knowledge change, and workflow execution are allowed only through declared authority boundaries, <b>PolicyDecision</b> outcomes, trace evidence, and recovery expectations.        |
| Known material and learning | A <b>KnowledgeRecord</b> is not factual truth by default; a <b>KnowledgeCandidate</b> is reserved for pending knowledge-boundary change, while a <b>LearningCandidate</b> is an evaluated proposed module improvement. |

### 3.2.1 Extended Rationale

- **Data Sovereignty:**

The first design obligation of **MADRE** is to keep the user's data under the user's control. Contemporary AI services often make private life, professional work, organizational knowledge,

and creative material flow through remote systems by default. That default is not neutral. It concentrates power in the companies that own the infrastructure, the logs, the model services, the policy switches, and the billing relation.

**MADRE** refuses to make surveillance a prerequisite for assistance. Local-first operation is a security boundary, a political boundary, and an engineering discipline. Private conversation, module-owned known material, organizational records, and sensitive work must remain local unless an explicit policy, classification, minimization, and trace path says otherwise.

Internet access is treated as a risk boundary. Prompt injection, credential exposure, external action abuse, provider lock-in, and remote data retention are normal failure modes of cloud-first agentic systems; **MADRE** 's architecture must make those failures harder by default and visible when they cannot be eliminated.

- **Free Software Duty:**

**MADRE** is released under GPL-3.0 because source availability alone is not enough. The project aims to provide usable agentic infrastructure for people and organizations that cannot, or should not, depend on platform owners for every intelligent interface they build.

The current generation of AI models draws on a broad inheritance of human language, labor, research, art, software, and social knowledge. A public-interest response requires practical, inspectable infrastructure outside exclusive corporate channels.

**MADRE** supports developers and creators running useful systems on ordinary machines, inspecting decisions, replacing inference backends and authorized integrations, and sharing improvements without requiring platform permission.

- **Model Agnosticism:**

No single model, host, vendor, format, or prompt convention should define **MADRE** 's architecture. Models improve, fail, disappear, become expensive, change license terms, and carry different bias and safety profiles.

**MADRE** treats inference as a governed **ModelAction**. The runtime may choose different permitted **InferenceProfile** and **InferenceBackend** combinations without letting any model become authority over policy, knowledge, actions, or truth.

- **Resource Discipline:**

Local-first AI must remain useful on ordinary commercial devices and degrade intelligently when resources are scarce.

Optional Complex Agent lanes, task decomposition, caching, context minimization, repeated checks where they matter, and durable records make stronger reasoning selective rather than an unlimited default.

- **Knowledge Governance:**

Agentic systems become dangerous when they blur text, known material, retained interaction, instruction, evidence, and action. A conversation turn is not a verified fact, an **InferenceOutput** is not a source or authority, and an action result is not an instruction.

A **KnowledgeRecord** carries provenance, sensitivity, intended use, **truthLevel**, and **truthAuthority**; a **LearningCandidate** is an evaluated module-improvement proposal, not ordinary stored knowledge.

- **Distributed Reasoning:**

Interaction can remain coherent while module-owned agents defer hard work, retrieve permitted material, test hypotheses, revise plans, and learn from evaluated outcomes without treating language models as minds.

Every agent, **InferenceRecord**, action invocation, **KnowledgeRecord** change, and **LearningCandidate** must remain accountable to an explicit runtime boundary.

## 3.3 Product Design

### 3.3.1 Product Scope

**MADRE** provides a disciplined coordination layer underneath agentic applications. The kernel routes interaction, registers the Core Module, records delayed work through **RuntimeJournal**, coordinates bounded context and actions, records **PolicyDecision** outcomes, and exposes recovery semantics.

The primary audience is developers and organizations building private, domain-aware, extensible assistant systems. Applications built on **MADRE** may vary in interface, domain, and policy, but they inherit the same authority model: modules own their agents and known material, model execution is constrained by **ModelBinding**, and policy, trace, action, and recovery boundaries remain runtime-controlled.

### 3.3.2 What MADRE Is

- A local-first runtime kernel for agentic systems.
- A model-agnostic coordination layer for multiple model families.
- A delayed-reasoning architecture that keeps conversation responsive while deeper work happens through module-local agents and **ReasoningTask** lifecycle.
- A policy-aware boundary for context, internal actions, knowledge, learning, and remote transfer.
- A developer foundation for building assistants, domain systems, and workflow-capable applications on top of governed local intelligence.

### 3.3.3 What MADRE Is Not

- **MADRE** is not a chatbot product. Chat is one possible surface over the kernel, not the kernel's definition.
- **MADRE** is not a prompt wrapper. Prompts are runtime inputs, not an architecture.
- **MADRE** is not a vendor abstraction layer whose purpose is merely to swap API providers.
- **MADRE** is not an unrestricted automation agent. Internal actions, external actions, knowledge-boundary change, and learning require explicit boundaries.
- **MADRE** is not a passive record of interaction history. The product is the governed runtime kernel and its architectural obligations.

### 3.3.4 Core User Value

**MADRE** lets an application answer quickly without pretending every answer can be safe, complete, or well-reasoned in the foreground. The Core Module can assign a **MainAgent** with a Complex Agent profile whose foreground lane clarifies intent, uses appropriately classified material, names uncertainty, and triggers deeper work. Module-local reasoning lanes perform work requiring decomposition, evidence, internal actions, verification, or stronger context.

This separation gives users a better contract. They are not forced to choose between a quick hallucination and a slow black box. They get immediate conversation when that is appropriate, delayed reasoning when that is necessary, and inspectable boundaries when the system touches sensitive data or meaningful actions.

### 3.3.5 Design Principles

- **Local-first, not local-only:** Local execution is the default because privacy, cost, resilience, and agency require a strong local base. Remote services may exist only as explicitly governed integrations with classification, minimization, policy, provenance, and journaled trace.
- **Model-agnostic by contract:** Models are replaceable runtime functions. The kernel must not let any model provider define the system's truth model, policy boundary, knowledge representation, or action authority.
- **Foreground lane, durable reasoning:** A **MainAgent** may use a Complex Agent foreground lane to preserve conversational flow. Expensive, uncertain, risky, or evidence-heavy work should become a scheduled **ReasoningTask** that can be resumed, inspected, verified, and delivered asynchronously.
- **Context is a governed asset:** Context is not just text stuffed into a prompt. It is selected, minimized, classified, sourced, scoped, and revoked according to the purpose of the task and the authority of the actor using it.
- **Autonomy is bounded:** **MADRE** must support unattended delayed work, but autonomy is legitimate only when the action has an applicable **PolicyDecision**, is recorded through **RuntimeEvent**, is recoverable for its risk level, and is blocked from crossing stronger boundaries without authorization.
- **Learning is evaluated:** Useful systems should improve from use, but a **LearningCandidate** cannot quietly rewrite protected behavior. Saving a **KnowledgeRecord** is not learning; knowledge-boundary and behavior changes must be reviewable, promotable, reversible, and separable.
- **Interfaces do not own the architecture:** The **madre** CLI, local read-first inspection console, graphical UI, speech surfaces, or domain applications are access surfaces. They must expose and operate the kernel without becoming the source of runtime authority.

### 3.3.6 Product Boundaries

The product boundary covers runtime responsibilities and architecture obligations: local-first operation, model-agnostic inference, module-owned agent coordination, governed context, bounded internal actions, knowledge handling, evaluated learning, runtime traceability, and recovery.

**MADRE** may expose many application forms: personal assistants, research systems, developer environments, organizational knowledge workers, or domain-specific automation environments. Those applications may differ in interface and policy, but they must inherit the same kernel commitments: local-first

operation, replaceable inference, governed context, bounded internal actions, inspectable **ConversationRecord** and **KnowledgeRecord** handling, evaluated learning, journaled runtime history, and recovery.



# Operational Scenarios

## 4.1 Scenario Cards

| Scenario trigger                                         | Module(s)                       | Runtime objects                                                          | Authority boundary                                                                                                       | Expected records                                                                               | Negative path                                                                                                       |
|----------------------------------------------------------|---------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Foreground continuity / ordinary interaction             | Core Module                     | MainAgent, ContextBundle, ConversationRecord                             | Answer only from permitted context and without making known material factual by implication.                             | Request and response RuntimeEvent; retained ConversationRecord where configured.               | Ambiguity, insufficient context, or risk creates clarification, block, or ReasoningTask, not an unsupported answer. |
| Delayed research evidence required                       | Core and research module        | ReasoningPlan, ReasoningTask, ContextBundle, optional ModelAction        | Routing, data scope, and each action require applicable PolicyDecision.                                                  | Work lifecycle and evidence references in RuntimeJournal; InferenceRecord when inference runs. | Unverified or disallowed sources do not become accepted knowledge or authoritative response.                        |
| Knowledge handling / material selected for module use    | Owning reasoning module         | KnowledgeRecord; KnowledgeCandidate                                      | Provenance, scope, sensitivity, intended use, truthLevel, truthAuthority, correction, and lifecycle metadata govern use. | Knowledge creation/correction/detachment RuntimeEvent and related PolicyDecision.              | Known material with unsuitable truth metadata is not presented as factual authority; saving it is not learning.     |
| Specialized interaction / domain assistance requested    | Core and selected domain module | ConversationRecord, ContextBundle, module-owned MADREAgent, ModelBinding | A transferred or delegated scope does not transfer policy or knowledge authority.                                        | Routing event, selected module, binding reference, context reference, and policy outcome.      | Domain-specific output remains InferenceOutput until explicitly handled; it does not self-route or self-promote.    |
| Workflow execution / reusable behavior invoked           | Owning reasoning module         | Workflow, AgentRoutine, AgentAction, ReasoningTask                       | Each executable action and external crossing is policy-gated with recovery expectation.                                  | Action, work, policy, failure, and recovery RuntimeEvent history.                              | A hidden prompt chain or unauthorized side effect is blocked or authorization-required.                             |
| Evaluated improvement / repeated outcome suggests change | Owning reasoning module         | LearningCandidate, evaluation evidence, optional KnowledgeRecord input   | Promotion changes module behavior only after declared evaluation and promotion authority.                                | Candidate creation, evaluation, promotion/rejection, and rollback events.                      | InferenceOutput, conversation retention, or KnowledgeRecord storage alone never activates learning.                 |

Developers may add models, internal actions, authorized external actions, projections, context policies, workflows, interfaces, and analysis helpers without rewriting the kernel, provided module ownership, ModelBinding, PolicyDecision, and RuntimeJournal trace obligations remain intact.

# Conceptual Architecture

## 5.1 Formal Requirements

This chapter states architectural requirements at product level.

| Requirement                       | Architectural meaning                                                                                                                                                                                                                   |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Local-first operation</b>      | The kernel must be able to run useful workflows on local user-owned infrastructure before optional remote integrations are considered.                                                                                                  |
| <b>Model agnosticism</b>          | All model execution enters through <code>ModelAction</code> , allowed <code>ModelBinding</code> , replaceable <code>InferenceProfile</code> , and <code>InferenceBackend</code> contracts, never through a vendor-specific core design. |
| <b>Delayed reasoning</b>          | The runtime must separate immediate conversation from deeper work that needs decomposition, evidence, verification, internal actions, or more compute.                                                                                  |
| <b>Context governance</b>         | Context must be selected, minimized, classified, provenanced, scoped, and revocable before it reaches models or internal actions.                                                                                                       |
| <b>Bounded internal actions</b>   | Executable behavior must be declared, policy-checked, traceable, and constrained by side effect and recovery rules. Typed runtime-defined internal actions are the canonical low-risk action form.                                      |
| <b>Record/knowledge isolation</b> | <code>ConversationRecord</code> , <code>KnowledgeRecord</code> , untrusted source material, <code>KnowledgeCandidate</code> , and <code>LearningCandidate</code> must not collapse into one undifferentiated prompt store.              |
| <b>Learning control</b>           | A <code>LearningCandidate</code> is an evaluated proposed improvement and changes active module behavior only through a defined promotion boundary; storing a <code>KnowledgeRecord</code> is not learning.                             |
| <b>Inspectable execution</b>      | The CLI, local developer console, <code>RuntimeJournal</code> , and read models must make runtime behavior understandable without bypassing policy.                                                                                     |

## 5.2 High-Level Architecture

**MADRE** is organized around a deterministic runtime kernel that coordinates governed reasoning modules through responsibility boundaries rather than through a fixed topology.

The central architectural rule is simple: the model is never the system. Models are invoked through **ModelAction** under context, policy, action, knowledge, and trace constraints. Each concrete **MADREAgent** is owned by one **ReasoningModule** and constrained by **ModelBinding**. Agents may be short-lived, persistent, specialized, conservative, speculative, or workflow-bound, but they do not own the kernel or carry module knowledge across ownership boundaries.

The kernel coordinates modules; it does not become the reasoning actor for every task. System-level reasoning belongs in governed modules. Kernel registration assigns one compatible **ReasoningModule** the Core Module role for incoming interaction, continuity, clarification, planning, delegation, and delayed-work decisions. Within that module, one **ComplexAgent** holds the **MainAgent** role; other **ComplexAgent** objects may exist without holding that role.

### 5.2.1 Conceptual Components

| Component                    | Responsibility                                                                                                                                                                                                                              |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Access surfaces              | Present the system through CLI, local read-first inspection console, graphical UI, speech, or domain applications without becoming the source of runtime authority.                                                                         |
| Core Module                  | A kernel-assigned <b>ReasoningModule</b> role that maintains interaction continuity, clarifies intent, creates reasoning plans, delegates to modules, and decides when deeper work is needed.                                               |
| Runtime kernel               | Coordinates module registration, routing, scheduling, resource limits, policies, action registries, durable records, recovery, and lifecycle.                                                                                               |
| Reasoning modules            | Execute bounded reasoning domains through module-owned agents, workflows, routines, actions, records, policies, known-material boundaries, and allowed model/inference scope.                                                               |
| Context and knowledge        | Provides scoped context bundles and <b>KnowledgeRecord</b> objects with provenance, scope, sensitivity, intended use, truth metadata, and correction paths.                                                                                 |
| Local Model Runtime Boundary | Product-level boundary for local or explicitly authorized inference through <b>ModelBinding</b> , <b>InferenceProfile</b> , <b>InferenceBackend</b> , <b>InferenceRecord</b> , and <b>InferenceOutput</b> .                                 |
| Internal Action Registry     | Exposes typed, bounded internal actions with declared inputs, outputs, side-effect class, policy requirements, trace evidence, and recovery expectations. External actions are exceptional risk boundaries, not the default execution path. |
| Learning and evolution       | Evaluates <b>LearningCandidate</b> objects and promotes only authorized module improvements; known-material storage is separate.                                                                                                            |
| Observability and recovery   | Gives developers and authorized users a read-first view of <b>ReasoningTask</b> , <b>RuntimeEvent</b> , <b>PolicyDecision</b> , failures, and rollback options through <b>RuntimeJournal</b> .                                              |

### 5.2.2 Responsibility Layer Diagram

Figure 5.1 presents the primary responsibility structure of **MADRE**. The diagram is not a sequential request pipeline. It describes the ownership boundaries that constrain all runtime behavior.

The **MADRE** runtime kernel is the deterministic system coordinator. It owns access-surface entry, module registration, Core Module assignment, module routing, scheduling, resource assignment, global policy boundaries, **RuntimeJournal** coordination, durable runtime state, and lifecycle boundaries. The kernel does not become the reasoning actor for ordinary tasks. It selects, activates, limits, and observes reasoning modules.

Reasoning modules are the runtime units that perform domain-bounded reasoning. The module assigned the Core Module role is the first receiver for ordinary user interaction because its registered capability covers interaction continuity, clarification, general planning, and delegation. Other modules may specialize in research, knowledge handling, code, media, resource optimization, or domain-specific work. A module may answer quickly, create delayed work, invoke internal actions, or evaluate outcomes when its local policy and assigned resources allow it.

Executable behavior is not exposed as arbitrary external tools. The normal execution path uses an Internal Action Registry composed of typed runtime-defined functions with declared inputs, outputs, side-effect class, policy requirements, `RuntimeEvent` trace obligations, and recovery expectations. The first reference implementation may implement these actions on a JVM runtime, but JVM binding is not an architecture-level authority. External services, host commands, provider-backed tools, remote model services, or arbitrary plugin surfaces are not part of the default path and require stronger authority boundaries.

The Local Model Runtime Boundary is useful product-level phrasing for the point below the module and action layer where allowed inference executes. It is realized through `ModelBinding`, `InferenceProfile`, `InferenceBackend`, `InferenceRecord`, and `InferenceOutput`; it is not a chatbot server or an authority. Generated material returns to the module as `InferenceOutput` and is not truth, policy, memory, knowledge, behavior, routing, authority, or a candidate by itself.

### 5.2.3 Concurrent Agent Lanes

**MADRE** separates immediate interaction from deeper reasoning, but this separation is not modeled as two global subsystems. It is a possible execution pattern for a module-owned `MADREAgent`.

A `MADREAgent` with a `ComplexAgent` profile may maintain three concurrent lanes. The foreground lane preserves conversational continuity, answers only when safe known context is sufficient, asks for clarification when intent or risk is unclear, and creates deferred work when more evidence is required. The reasoning lane performs planning, decomposition, evidence gathering, action invocation, verification, and result construction under module authority. The reflection lane reviews recent work, observes delayed outcomes, detects failure patterns, and proposes evaluated `LearningCandidate` objects.

This pattern may exist in any reasoning module. The Core Module may use it for general interaction and delegation. A research module may use it for evidence comparison and verification. A knowledge module may use it for extraction, provenance checks, correction, and deletion semantics. The kernel schedules and constrains these lanes through module routing, resource limits, durable state, policy, and `RuntimeJournal`, but the reasoning behavior itself belongs to module-local agents.

## 5.3 Runtime Data Flow

Runtime data flow in **MADRE** is concurrent, governed, and module-centered. A local request enters through an access surface and is submitted to the kernel. The kernel records the request, applies the relevant entry policy, and routes it to the Core Module by registration or to another registered module when routing evidence supports that decision.

Within the selected module, its `MainAgent` may produce an immediate foreground response, create or continue delayed reasoning work, invoke internal actions, request inference through an allowed `ModelAction`, and reflect on recent outcomes. These activities are coordinated inside a module-owned agent and constrained by module policy, kernel policy, `ModelBinding`, resource assignment, durable state, `RuntimeJournal`, and recovery rules.

The following subsections describe the canonical request narrative, the governed model/action sequence, and the data-authority rules that prevent user input, source material, `InferenceOutput`, action

results, or learning candidates from promoting themselves into authority-bearing state.

### 5.3.1 Canonical Request Narrative

The canonical user request begins when an access surface submits a local request to the kernel. The kernel does not interpret the request as a **KnowledgeRecord**, factual authority, or policy. It records a **RuntimeEvent**, applies entry constraints, and routes the work to the Core Module unless another module is already selected by session state, policy, or module registry evidence.

The receiving module evaluates the request through its module-local **MainAgent**. If a configured foreground lane has enough safe local context, it may produce an immediate answer. If intent, scope, or risk is unclear, it asks for clarification. If the request requires evidence, stronger reasoning, internal actions, model inference, or broader context, the reasoning lane creates or continues a **ReasoningTask**. Reflection may run after the response or after delayed outcomes to propose a **LearningCandidate**. Knowledge-boundary change is separately represented as a **KnowledgeCandidate**.

Every significant step emits a **RuntimeEvent** through **RuntimeJournal**. Context bundles, actions, **InferenceRecord** and **InferenceOutput** objects, **PolicyDecision** outcomes, failures, correction, roll-back, invalidation, supersession, and deletion/detachment remain inspectable through read-first surfaces.

### 5.3.2 Governed Model And Action Sequence

When module work requires executable behavior, the module invokes an internal typed action. The default action mechanism is local, typed, runtime-defined, policy-gated, and traceable. It is not an open plugin socket and not a network-exposed external action protocol.

A **ModelAction** is an **AgentAction** that uses inference. Before it runs, the module supplies a governed **ContextBundle**, an allowed **ModelBinding** and **InferenceProfile**, and the applicable **PolicyDecision**. An **InferenceBackend** executes inference without allowing a provider, template, or runtime format to define **MADRE**'s authority model. One execution is captured as an **InferenceRecord**; its generated material is an **InferenceOutput**.

Action results may be observations. **InferenceOutput** is generated material and is not a candidate by itself. It may inform a next reasoning step or undergo explicit handling into another record, but it does not authorize actions, create policy, become memory or knowledge, change routing or behavior, or promote learning. Each such transition requires its relevant authority boundary, verification path, **RuntimeEvent** record, and recovery expectation.

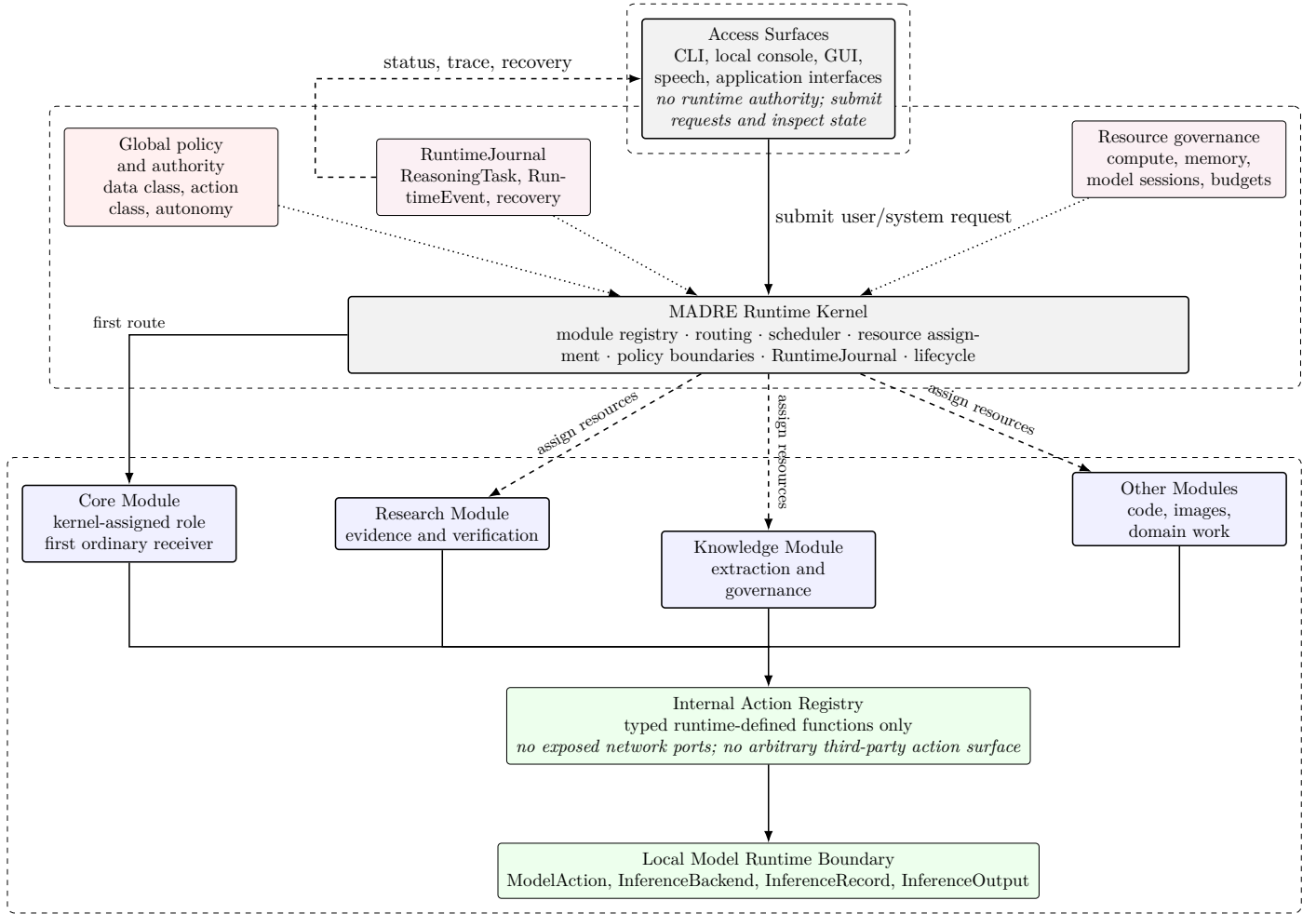


Figure 5.1: MADRE responsibility layers: kernel coordination, Core Module assignment, runtime journal, actions, and inference.

Figure 5.1 defines the top-level responsibility model. Access surfaces are intentionally outside runtime authority: they submit requests and inspect state, but they do not own policy, knowledge, routing, scheduling, or action-authority decisions. The kernel is the only system-level coordinator. It registers modules, assigns the Core Module, routes requests, assigns resources, applies global policy boundaries, records through `RuntimeJournal`, and maintains durable lifecycle state.

The module layer is the reasoning layer. The Core Module is shown first because ordinary interaction is routed there by registration, not because it is outside the module system. Research, knowledge, code, media, or domain modules are peers selected when the registry, policies, and current reasoning plan indicate a better fit. Internal actions and inference appear below the module layer because they are invoked by authorized module behavior, not by access surfaces or `InferenceOutput` directly.

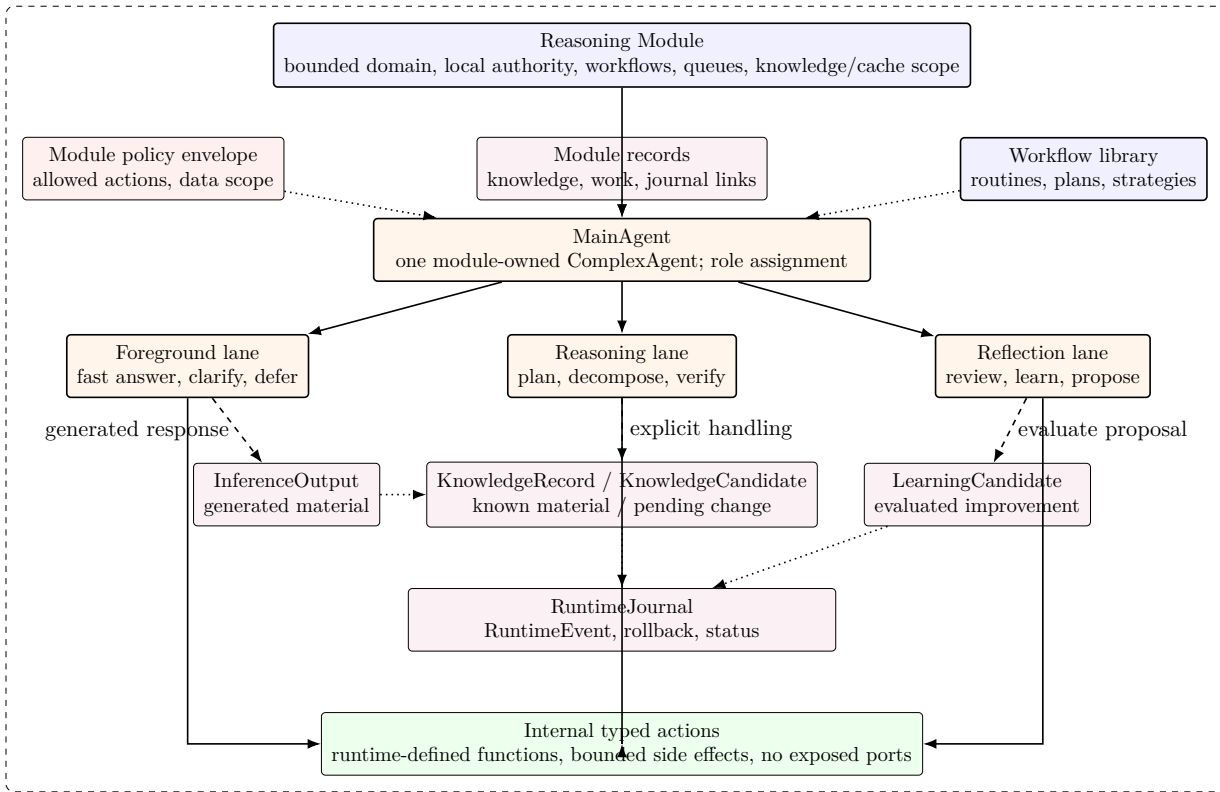


Figure 5.2: Reasoning module internal structure with its MainAgent using a ComplexAgent lane profile.

Figure 5.2 expands the internal structure of a reasoning module. A module owns a bounded domain, local policy envelope, records, workflow library, knowledge boundary, and concrete **MADREAgent** objects. One module-owned **ComplexAgent** holds the **MainAgent** role; additional **ComplexAgent** objects may serve other module work without that role.

The foreground lane, reasoning lane, and reflection lane may operate concurrently. The foreground lane protects user continuity. The reasoning lane protects evidence, decomposition, verification, and stronger work. The reflection lane protects adaptation by evaluating a proposed **LearningCandidate** rather than silently rewriting runtime behavior. These lanes do not increase agent authority. They remain constrained by module policy, kernel policy, assigned resources, trace requirements, and promotion boundaries.



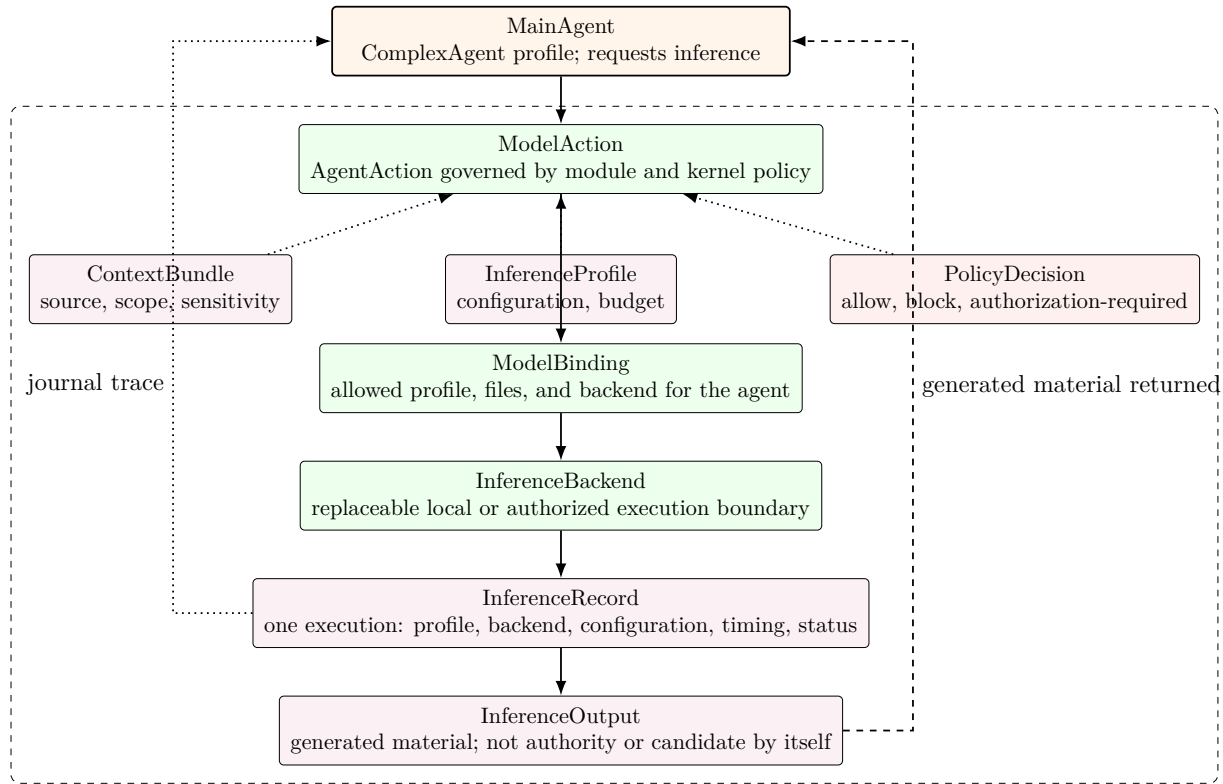


Figure 5.3: ModelAction inference path from governed execution through InferenceRecord to non-authoritative InferenceOutput.

Figure 5.3 describes inference as the governed ModelAction path rather than as an external server call or chat-completion product boundary. A module-owned MADREAgent, including one serving as MainAgent, can request generation, critique, classification, or verification only under its allowed ModelBinding and an applicable PolicyDecision.

The InferenceProfile captures execution configuration, and the replaceable InferenceBackend executes the permitted inference. Every execution creates an InferenceRecord; the generated material attached to that execution is InferenceOutput. The Local Model Runtime Boundary remains useful as product wording for this controlled local or authorized execution boundary, not as a competing runtime class. InferenceOutput is not truth, policy, memory, knowledge, behavior, routing, authority, or a candidate by itself. Explicit module handling can retain it as ConversationRecord, save appropriately classified material as KnowledgeRecord, evaluate it toward a LearningCandidate, or discard it, with the transition recorded in RuntimeJournal.

### 5.3.3 Data And Authority Rules

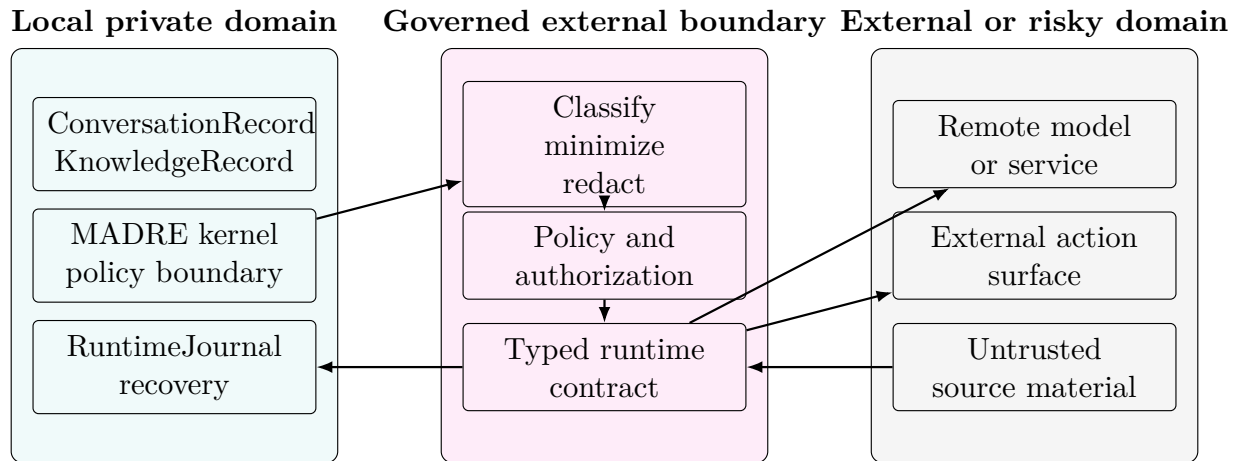
- User input enters as ConversationRecord material when retained, not as policy, knowledge, truth, action authority, or model authority.
- Retrieved, extracted, or saved material becomes KnowledgeRecord only with provenance, scope, sensitivity, intended use, truthLevel, truthAuthority, correction, and lifecycle metadata; it is not factual truth by default.
- Context bundles are constructed, minimized, classified, scoped, and revocable before they reach an agent lane, internal action, or ModelAction.

- **InferenceOutput** is generated material from an **InferenceRecord**; it does not become an answer, policy decision, memory, knowledge, behavior, routing, authority, or candidate by itself.
- Internal action results enter as bounded observations. They become relied-on evidence only according to the action contract, verification path, and **RuntimeEvent** trace.
- External action surfaces, host commands, remote services, arbitrary filesystem mutation, credentials, and irreversible effects require stronger authority boundaries than ordinary internal actions.
- A **KnowledgeCandidate** represents only pending knowledge-boundary change; a **LearningCandidate** represents evaluated proposed module improvement and cannot directly rewrite active runtime behavior.

### 5.3.4 Trust Boundary Diagram

The trust boundary diagram distinguishes the ordinary local governed domain from optional external or risky domains. The default execution path remains inside the local governed domain: access surfaces, kernel coordination, reasoning modules, internal typed actions, local model runtime, **RuntimeJournal**, and recovery.

The external boundary exists for exceptional external actions: remote services, provider-backed models, host-command execution, arbitrary filesystem mutation, network access, untrusted source acquisition, or third-party integrations. These are not ordinary internal actions. They require classification, minimization, explicit policy authorization, recorded trace, and a recovery or blocking expectation appropriate to their risk class.



### 5.3.5 Access Surfaces

**MADRE** exposes access surfaces without allowing those surfaces to define the kernel. The **madre** CLI submits local requests and exposes status or result evidence. A local read-first inspection console may expose sessions, queued work, context bundles, policy outcomes, model evidence, internal action evidence, **KnowledgeCandidate**, **LearningCandidate**, **RuntimeJournal** traces, recovery, and runtime state. These surfaces operate under the same policy and authority model as every other runtime interaction.

The inspection console is not a privileged mutation path. Its transport is intentionally not fixed at architecture level. If a later implementation uses HTTP, sockets, embedded UI, or another mechanism, that choice must be justified by deployment and security decisions and must preserve local-first, read-first, policy-governed behavior.

Graphical, speech, product-specific, or integration surfaces remain access surfaces over the runtime rather than privileged paths around it.

# Safety And Evolution Policy

## 6.1 Runtime Safety

**MADRE** is intended to run useful delayed work while the user is not constantly present. That requirement makes safety an architectural property, not a documentation afterthought. The system must be able to explain what it did, why it did it, what data it used, what boundary allowed it, and how the result can be corrected or rolled back when the action is reversible.

### 6.1.1 Policy Before Action

The kernel must not expose executable behavior merely because a model can request it. Internal actions, external calls, knowledge-boundary changes, workflow execution, and learning promotion need an explicit `PolicyDecision` before execution. The safe path should be the normal path, and safe autonomous improvement should not require constant user interruption. Unsafe paths should be technically isolated, traceable, recoverable where possible, and visible to the responsible developer or operator through `RuntimeJournal`.

Typed runtime-defined internal actions represent the canonical bounded action form. They may be allowed only through a policy-gated action boundary. Remote action, host commands, arbitrary filesystem access, mutation actions, provider-backed integrations, active learning promotion, and workflow mutation require stronger authority boundaries than bounded internal action execution.

### 6.1.2 Autonomy Levels

| Level            | Meaning                                                                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Observation      | Read local runtime state, existing knowledge, and safe metadata without mutation.                                                                       |
| Draft creation   | Produce summaries, candidate records, or proposed changes without promoting them into active state.                                                     |
| Bounded mutation | Modify authorized local state only when the action is logged, scoped, reversible where possible, and policy-permitted.                                  |
| External action  | Contact remote systems, use external services, or disclose data only after classification, minimization, authorization, provenance, and recorded trace. |
| Critical action  | Credentials, identity, legal, financial, destructive, irreversible, or executable-code actions require a stronger human authorization path.             |

## 6.2 Evolution Control

### 6.2.1 Evolution Rules

- Runtime learning is represented by evaluated **LearningCandidate** proposals, not only by manual feedback collection.
- Learning strategies must declare what they are allowed to observe, propose, mutate, verify, and promote.
- A **LearningCandidate** cannot silently modify protected behavior.
- Learning proposals begin with source, evaluation evidence, scope, and rollback information before they are promoted.
- A **KnowledgeCandidate** is used only for pending knowledge-boundary change and requires provenance, truth metadata, correction paths, and recovery options.
- Saving a **KnowledgeRecord** is not learning by itself.
- Workflow and routing improvements may become default behavior through developer-defined checks, evaluation gates, and rollback policies, without requiring user authorization for every routine improvement.
- Model adaptation, fine-tuning, or weight updates are advanced model-evolution actions governed by stronger model-adaptation authority.
- Critical learning failures must be recoverable through snapshots, versioned state, VCS-backed records, or an equivalent rollback mechanism.

This policy keeps **MADRE** capable of evolving through conscious, checked learning without reducing improvement to constant user authorization. The final user should be interrupted as little as possible. Routine learning should be governed by developer-defined strategies, evidence, tests, promotion gates, **RuntimeJournal** traces, and rollback mechanisms; user authorization is reserved for boundaries where risk, privacy, identity, external action, or policy explicitly requires it.

D-011 fixes the controlled-promotion rule: **KnowledgeCandidate** denotes pending knowledge-boundary change and **LearningCandidate** denotes evaluated proposed module improvement. Neither candidate creation nor **KnowledgeRecord** storage silently rewrites protected behavior.

# System Engineering Baseline

## 7.1 Engineering Problem

**MADRE** addresses the engineering problem of coordinating model inference without allowing the model to become the system. The runtime must support foreground continuity, delayed reasoning, governed context, bounded internal actions, known material, learning, runtime traceability, and recovery while preserving local ownership of data and operational control.

The core difficulty is not only model accuracy. A useful local assistant must decide when an answer is safe in the foreground lane, when work must be delayed, which context may be exposed, which internal action may be invoked, which result requires verification, and how a mistaken or unsafe outcome can be inspected, corrected, or rolled back.

## 7.2 Stakeholders And Concerns

**MADRE** serves users who need private assistance, developers who need an extensible runtime, organizations that need governed local knowledge, and maintainers who must keep the system auditable and evolvable. Each stakeholder depends on the same architectural guarantees: model independence, local-first operation, explicit authority boundaries, understandable execution, and recoverable state.

The principal concerns are privacy, cost, autonomy control, model substitution, context minimization, data provenance, action safety, knowledge correction, learning control, operational continuity, and evidence for system behavior. A runtime that cannot expose those concerns as inspectable state cannot satisfy **MADRE**'s product objective, even if its conversational output appears fluent.

## 7.3 Requirement Domains

**MADRE**'s functional baseline is organized around runtime domains. The foreground lane domain preserves conversational flow, asks for clarification, and creates deeper work when the answer requires evidence or internal actions. The delayed reasoning domain records work durably, schedules it, verifies results, and returns outcomes without erasing the foreground interaction.

The context domain selects, minimizes, classifies, scopes, and revokes information before it reaches models or internal actions. The inference domain uses `ModelAction`, permitted `ModelBinding`, `InferenceProfile`, and replaceable `InferenceBackend` contracts. The internal action domain exposes declared side effects under `PolicyDecision` and `RuntimeEvent` recording. The knowledge domain separates `ConversationRecord`, `KnowledgeRecord`, pending `KnowledgeCandidate`, untrusted source material, and evaluated `LearningCandidate`.

The learning domain keeps improvement proposals quarantined until evidence, evaluation, promotion, and rollback rules allow activation. The observability domain exposes sessions, `ReasoningTask` lifecycle, context bundles, `InferenceRecord`, actions, `PolicyDecision`, failures, metrics, and recovery paths. The deployment domain keeps ordinary local operation viable before optional remote integrations are considered.

## 7.4 Traceability Surface

Each significant product claim in **MADRE** must be visible as system behavior. A claim such as local-first operation corresponds to local execution paths, storage boundaries, remote-transfer controls, and inspection evidence. A claim such as model agnosticism corresponds to replaceable **InferenceProfile** and **InferenceBackend** contracts and tests that prevent provider-specific assumptions from entering the kernel.

The same trace applies to safety claims. Policy before action must appear as an enforceable boundary around internal actions, external calls, knowledge-boundary change, workflow execution, and learning promotion. Governed context must appear as provenance, classification, minimization, scope, revocation, and journaled trace. Delayed reasoning must appear as **ReasoningTask**, **RuntimeEvent** transitions, verification, result delivery, and recovery.

## 7.5 Quality Attributes

**MADRE** prioritizes data sovereignty, privacy, resource discipline, responsiveness, reliability, recoverability, maintainability, portability, auditability, and resistance to hallucination-driven action. These attributes are not independent decorations; they define whether the runtime fulfills its purpose.

Responsiveness is measured by preserving useful foreground conversation while heavier work is delayed. Reliability is measured by durable records, repeatable status, bounded retries, and visible failures. Maintainability is measured by the ability to replace models, internal actions, workflows, and interfaces without rewriting the kernel. Auditability is measured by the ability to explain what context, **InferenceRecord**, action, **PolicyDecision**, and verification path produced an outcome through **RuntimeJournal**.

## 7.6 Risk And Policy Boundaries

**MADRE** treats prompt injection, data leakage, excessive agency, unsafe action execution, poisoned knowledge, provider dependency, remote retention, unbounded consumption, credential exposure, and rollback gaps as architectural risks. The runtime must assume that **InferenceOutput** can be persuasive, wrong, incomplete, unsafe, or maliciously influenced by context.

Risk control is expressed through policy boundaries. Data classification governs what may be retrieved, stored, disclosed, or sent to a model. Autonomy level governs whether a result may remain a draft, update local state, invoke an internal action, contact an external service, or require stronger authorization. Recovery expectation governs whether the action must be reversible, snapshot-backed, compensatable, or blocked.

# Requirements Baseline

## 8.1 Requirement Taxonomy

| Prefix | Requirement family                                                                                                                                                           |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR     | Functional behavior of foreground-lane conversation, delayed reasoning, context construction, inference, internal actions, knowledge, learning, runtime trace, and recovery. |
| NFR    | Quality attributes such as responsiveness, reliability, recoverability, maintainability, portability, resource discipline, and auditability.                                 |
| SEC    | Security and trust controls, including prompt injection resistance, credential protection, action authority, and remote-transfer boundaries.                                 |
| DATA   | Data, knowledge, provenance, retention, deletion, correction, revocation, and quarantine requirements.                                                                       |
| AI     | <b>ModelAction</b> , model substitution, <b>InferenceOutput</b> handling, and verification requirements.                                                                     |
| UX     | User-visible states, clarification, authorization, progress, failure, correction, and result-delivery requirements.                                                          |
| OPS    | Local deployment, storage, backup, diagnostics, migration, and operational recovery requirements.                                                                            |



## 8.2 Functional Requirements

| ID     | Requirement                                                                                                                                                                                                                         | Acceptance criteria                                                                                                                                                                      | Verification                                                        |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| FR-001 | The runtime shall preserve foreground-lane conversation while delaying evidence-heavy, risky, action-dependent, or broad-context work.                                                                                              | Immediate answers stay within safe known context; stronger work is recorded as delayed work.                                                                                             | Triage scenarios; D-001, D-004.                                     |
| FR-002 | The runtime shall represent delayed work as <b>ReasoningTask</b> objects with inspectable status and recovery path.                                                                                                                 | Work exposes queued, running, blocked, failed, completed, and recovered states with <b>RuntimeEvent</b> history in <b>RuntimeJournal</b> .                                               | Work-lifecycle and recovery scenarios; D-001, D-003, D-008.         |
| FR-003 | The runtime shall build context bundles from sourced, classified, minimized, scoped, provenanced, and revocable material.                                                                                                           | Context is validated or rejected before <b>ModelAction</b> or internal action use.                                                                                                       | Context-governance scenarios; D-001, D-005, D-007.                  |
| FR-004 | The runtime shall execute model-backed work through <b>ModelAction</b> constrained by <b>ModelBinding</b> and replaceable <b>InferenceProfile/InferenceBar</b> and its <b>InferenceRecord</b> and <b>InferenceOutput</b> contracts. | Every execution produces an inspectable <b>InferenceRecord</b> and keeps its <b>InferenceOutput</b> and <b>InferenceBar</b> alternative.                                                 | Inference substitution and output-boundary scenarios; D-001, D-006. |
| FR-005 | The runtime shall expose executable behavior only as bounded internal actions or explicitly authorized external actions.                                                                                                            | Internal actions remain policy-gated, while disallowed action requests are blocked or authorization-gated.                                                                               | Action policy scenarios; D-007, D-009.                              |
| FR-006 | The runtime shall keep <b>KnowledgeCandidate</b> and <b>LearningCandidate</b> semantics separate and inactive unless their respective promotion rules activate them.                                                                | Saving a <b>KnowledgeRecord</b> does not activate learning; evaluated behavior, routing, workflow, or inference-selection improvement remains a <b>LearningCandidate</b> until promoted. | Knowledge-boundary and learning-promotion scenarios; D-011.         |

| ID     | Requirement                                                             | Acceptance criteria                                                                                                        | Verification                                  |
|--------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| FR-007 | The runtime shall expose local request entry and read-first inspection. | Local CLI request/status interaction and local console inspection expose system state without bypassing runtime authority. | Local access and inspection scenarios; D-010. |
| FR-008 | The runtime shall represent recovery outcomes explicitly.               | Correction, rollback, deletion/detachment, invalidation, and supersession are distinguishable and auditable.               | Recovery readout scenarios; D-003, D-008.     |

### 8.3 Cross-Cutting Requirements

| ID      | Requirement                                                                                                                                                                        | Rationale                                                    | Verification                                       |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------|
| NFR-001 | Local useful operation shall remain possible without mandatory remote service dependency.                                                                                          | Data sovereignty, cost control, and resilience.              | Local execution evidence; D-001, D-002, D-006.     |
| NFR-002 | Runtime state changes shall be inspectable through local read models.                                                                                                              | Auditability and recoverability.                             | Inspection review; D-003, D-010.                   |
| NFR-003 | Delayed work shall be bounded by status, resource, and failure discipline.                                                                                                         | Prevents unbounded cost, compute, storage, and queue growth. | Resource/status scenarios; D-003, D-007, D-010.    |
| SEC-001 | Prompt or source material shall not become instruction, policy, or authority without explicit classification and boundary checks.                                                  | Source-as-instruction injection risk.                        | Adversarial context scenario; D-004, D-005, D-007. |
| SEC-002 | A <code>PolicyDecision</code> shall explicitly represent allow, block, or authorization-required before action execution or promotion and be linked to <code>RuntimeEvent</code> . | Prevents hidden authority decisions and excessive agency.    | Policy matrix review; D-007, D-009, D-011.         |

| ID       | Requirement                                                                                                                                                                                            | Rationale                                                             | Verification                                                                                 |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| SEC-003  | External, destructive, credential, legal, financial, executable, or irreversible effects shall be blocked or separately authorized.                                                                    | Preserves recoverability and explicit authority over critical action. | Critical-action negative paths; D-007, D-008, D-009.                                         |
| DATA-001 | A <b>KnowledgeRecord</b> shall carry provenance, scope, sensitivity, intended use, <b>truthLevel</b> , <b>truthAuthority</b> , correction, and lifecycle metadata; it is not factual truth by default. | Knowledge governance and user control.                                | Non-authoritative known-material, correction, and revocation scenarios; D-005, D-008, D-011. |
| DATA-002 | Active state and accountable <b>RuntimeEvent</b> history shall remain distinguishable.                                                                                                                 | Allows deletion/detachment without losing required trace history.     | Recovery inspection scenarios; D-003, D-008.                                                 |
| AI-001   | <b>InferenceOutput</b> shall remain generated material from an <b>InferenceRecord</b> , not truth, policy, memory, knowledge, behavior, routing, authority, or a candidate by itself.                  | Hallucination and false-authority risk.                               | Output non-authority and non-candidate scenarios; D-001, D-006.                              |
| AI-002   | Models and <b>InferenceOutput</b> shall not create or widen <b>PolicyDecision</b> , authorize actions, establish knowledge truth, alter behavior or routing, or activate learning.                     | Keeps runtime authority outside generated material.                   | Model-overreach and promotion-control scenarios; D-006, D-007, D-011.                        |
| UX-001   | Users shall be able to distinguish immediate answers, queued work, running work, blocked work, failed work, completed work, and authorization requests.                                                | Continuity and user control.                                          | Interaction-state review; D-004, D-010, D-011.                                               |

| ID      | Requirement                                                                                                                              | Rationale                                         | Verification                                 |
|---------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------------------------|
| OPS-001 | Local access surfaces shall submit requests and inspect <code>RuntimeJournal</code> status and trace without becoming runtime authority. | Supports operability while preserving governance. | Local access and inspection scenario; D-010. |

# Traceability Model

## 9.1 Traceability Method

Traceability in **MADRE** links the product claim, stakeholder concern, requirement, architecture obligation, validation method, and evidence status. The purpose is to keep evaluation anchored to the system thesis: local-first operation, model agnosticism, delayed reasoning, governed context, bounded internal actions, evaluated learning promotion, runtime traceability, and recovery.

## 9.2 Architecture Traceability Matrix

| Claim                               | Concern                            | Requirement      | Decision trace      | Evidence family                                                                                                                |
|-------------------------------------|------------------------------------|------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Local-first operation               | Data sovereignty, cost, resilience | NFR-001          | D-001, D-002, D-010 | Local request, local status, local journal trace, no mandatory remote service.                                                 |
| Model agnosticism                   | Provider independence              | FR-004, AI-001   | D-006               | Replaceable InferenceBackend/InferenceOutput substitution plus non-authority scenario.                                         |
| Delayed reasoning                   | Responsiveness and quality         | FR-001, FR-002   | D-001, D-003, D-004 | Foreground triage plus durable delayed-work lifecycle.                                                                         |
| Governed context and known material | Privacy and correctness            | FR-003, DATA-001 | D-005, D-007        | Context acceptance, source-as-data, non-authoritative KnowledgeRecord use, truth-metadata correction, and revocation scenario. |

| Claim                      | Concern                       | Requirement               | Decision trace      | Evidence family                                                                                                        |
|----------------------------|-------------------------------|---------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------|
| Bounded internal actions   | Safety and recoverability     | FR-005, SEC-001           | D-007, D-009        | Policy-gated internal action and blocked unsafe external action attempts.                                              |
| Learning promotion control | Controlled evolution          | FR-006                    | D-011               | LearningCandidate evaluation and inspection without active promotion; KnowledgeRecord save is not treated as learning. |
| Runtime trace and recovery | Accountability and correction | FR-008, NFR-002, DATA-002 | D-003, D-008, D-010 | RuntimeJournal/Runtime status, correction, invalidation, supersession, roll-back, and deletion/detachment trace.       |

### 9.3 Integrated Architecture Trace

| Step              | Required behavior                                                                       | Decision trace | Negative evidence                                           |
|-------------------|-----------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------|
| Local entry       | Request enters through local access surface.                                            | D-001, D-010   | Internal-only execution is not enough to prove inspection.  |
| Foreground triage | Foreground answers only safe known context, clarifies, blocks, or creates delayed work. | D-004, D-007   | Evidence-heavy request is not answered as if already known. |
| Durable work      | Delayed work is represented by ReasoningTask with RuntimeEvent status history.          | D-003          | Work is not lost or silently overwritten.                   |

| Step                            | Required behavior                                                                                                                                                 | Decision trace | Negative evidence                                                                                         |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------|
| Governed context                | Context is sourced, classified, minimized, scoped, provenanced, revocable, and linked to journal trace.                                                           | D-005, D-007   | Sensitive, unrelated, revoked, or incorrectly truth-classified material is excluded or blocked.           |
| Inference boundary              | <code>ModelAction</code> uses <code>InferenceBackend</code> , creates <code>InferenceRecord</code> , and returns non-authoritative <code>InferenceOutput</code> . | D-006          | Generated material does not become authority or a candidate by itself.                                    |
| Bounded internal action         | Typed runtime-defined internal actions are policy-gated and recorded.                                                                                             | D-009          | Remote, host-command, filesystem, or mutation actions require stronger authority boundaries.              |
| Knowledge and learning handling | <code>KnowledgeCandidate</code> covers pending knowledge-boundary change; <code>LearningCandidate</code> covers evaluated improvement.                            | D-011          | <code>KnowledgeRecord</code> storage does not activate learning and candidates cannot promote themselves. |
| Inspection and recovery         | Status, trace, failure, correction, rollback, invalidation, supersession, and deletion/detachment are visible through <code>RuntimeJournal</code> .               | D-008, D-010   | Active state changes do not erase accountable history.                                                    |

## Design Decision Traceability

Design decisions D-001 through D-011 provide traceability from product requirements to architecture obligations and validation evidence. They identify accepted product architecture decisions for governed delayed reasoning, durable work, context, inference, policy, recovery, bounded internal actions, local access, knowledge-boundary control, and evaluated learning promotion.

| Decision | Product-level effect                                                                                                            | Architecture boundary                                                                                                        |
|----------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| D-001    | Defines source-backed delayed-answer behavior as a core expression of governed runtime value.                                   | Conversation remains connected to durable work and evidence.                                                                 |
| D-002    | Establishes behavior-first architectural validation as part of the product's engineering discipline.                            | Observable behavior anchors architecture claims.                                                                             |
| D-003    | Requires <b>ReasoningTask</b> lifecycle and projected inspection state.                                                         | <b>RuntimeJournal</b> and append-only <b>RuntimeEvent</b> preserve local traceability.                                       |
| D-004    | Requires conservative foreground triage.                                                                                        | Unsupported answers become clarification, policy block, or delayed work.                                                     |
| D-005    | Requires governed context acceptance and rejection.                                                                             | Context is sourced, classified, minimized, scoped, provenanced, and revocable.                                               |
| D-006    | Requires <b>ModelAction</b> , <b>InferenceBackend</b> , <b>InferenceRecord</b> , and non-authoritative <b>InferenceOutput</b> . | Models remain runtime functions, not authorities or candidate creators by generation alone.                                  |
| D-007    | Requires explicit <b>PolicyDecision</b> before action execution.                                                                | Action execution follows policy rather than model instruction.                                                               |
| D-008    | Requires recovery as a recorded semantic transition.                                                                            | Correction, rollback, invalidation, supersession, deletion, and detachment have distinguishable <b>RuntimeEvent</b> meaning. |



| Decision | Product-level effect                                                                                               | Architecture boundary                                                                                                                                         |
|----------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D-009    | Defines bounded internal actions as the low-risk executable reference.                                             | Action execution is policy-gated, traceable, and recoverability-aware.                                                                                        |
| D-010    | Defines CLI request/status interaction and local read-first console inspection as canonical local access surfaces. | Access surfaces expose the kernel without becoming runtime authority.                                                                                         |
| D-011    | Distinguishes <b>KnowledgeCandidate</b> from <b>LearningCandidate</b> improvements.                                | pending boundary changes from evaluated module<br>Known-material storage is not learning; proposed evolution remains inspectable until promoted by authority. |

# Architecture Viewpoints

## 11.1 Viewpoint Catalogue

| Viewpoint       | Primary concern                  | MADRE content                                                                                                                                                                           |
|-----------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| System context  | Runtime boundary and environment | User, local device, filesystem, operating system, local models, optional external integrations, internal actions, and untrusted sources.                                                |
| Logical         | Responsibility separation        | Kernel, Core Module, module-owned agents, scheduler, context/knowledge, Local Model Runtime Boundary, Internal Action Registry, PolicyDecision, RuntimeJournal, learning, and recovery. |
| Runtime/process | Work lifecycle                   | User turn, triage, ReasoningTask, context bundle, scheduling, InferenceRecord, result delivery, RuntimeEvent, and recovery.                                                             |
| Data/knowledge  | Data ownership and provenance    | ConversationRecord, KnowledgeRecord, KnowledgeCandidate, LearningCandidate, truth metadata, retention, correction, and deletion.                                                        |
| Security/trust  | Boundary enforcement             | Local domain, governed external boundary, external/risky domain, prompt injection, remote transfer, credential exposure, and action side effects.                                       |

| Viewpoint  | Primary concern      | MADRE content                                                                                                                                                                   |
|------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Deployment | Local operation      | Ordinary device profile, local storage, model availability, backup, migration, diagnostics, and optional remote configuration.                                                  |
| Extension  | Developer evolution  | <code>ModelBinding</code> , <code>InferenceProfile</code> , <code>InferenceBackend</code> , action contracts, workflows, projections, context policies, and interface surfaces. |
| UX         | User-visible control | Conversation, clarification, queue state, progress, authorization, failure, completion, correction, and rollback.                                                               |

## 11.2 Viewpoint Design Focus

| Viewpoint       | Architecture focus                                                                                                                                     | Authority boundary                                                                                                                                                             |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| System context  | Local user, local device, CLI, local read-first console, local sources, and optional remote integrations.                                              | Local authority remains explicit across every access surface.                                                                                                                  |
| Logical         | Kernel coordination over routing, delayed work, context, bounded action use, <code>PolicyDecision</code> , <code>RuntimeJournal</code> , and recovery. | Runtime authority is not delegated to interfaces, models, actions, or <code>InferenceOutput</code> .                                                                           |
| Runtime/process | Governed lifecycle from local request to inspection and recovery.                                                                                      | <code>ReasoningTask</code> transitions remain recorded, inspectable, and recoverable.                                                                                          |
| Data/knowledge  | Governed context and explicit knowledge/learning types.                                                                                                | Source material, <code>ConversationRecord</code> , <code>KnowledgeRecord</code> , <code>KnowledgeCandidate</code> , and <code>LearningCandidate</code> remain distinguishable. |

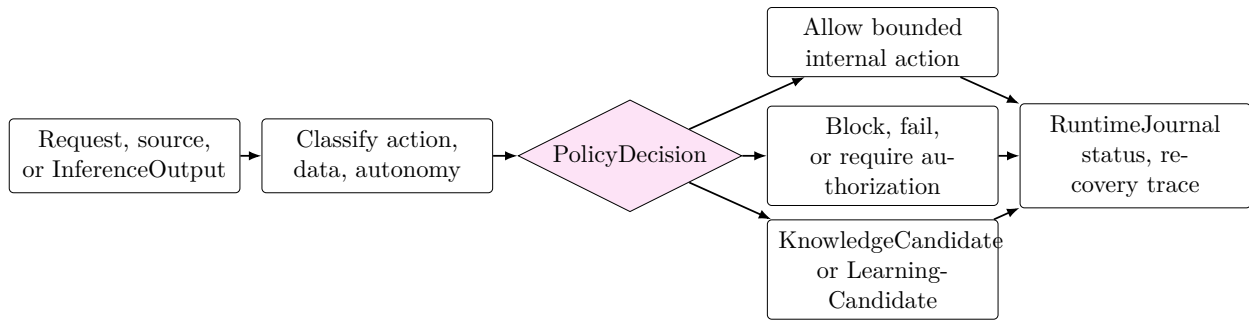
| Viewpoint              | Architecture focus                                                                                                      | Authority boundary                                                |
|------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Security/trust         | Source-as-data, model-as-function, policy before action, and bounded internal action authority.                         | Action execution follows declared policy outcomes.                |
| Observability/recovery | Local status, <code>RuntimeJournal</code> , verification, policy, knowledge/learning transition, and recovery evidence. | Read-first inspection must not become a privileged mutation path. |
| Extension              | Inference backends, actions, workflows, and learning systems inherit the defined boundaries.                            | Extension preserves kernel authority and product traceability.    |

## Runtime Governance

### 12.1 Runtime Authority Rule

Runtime authority belongs to the **MADRE** kernel and its policy boundaries. User turns, source text, **InferenceOutput**, action results, local inspection, **KnowledgeCandidate**, and **LearningCandidate** can inform decisions, but they cannot authorize their own authority, promote themselves, mutate protected state, or widen permissions. Bounded internal action use requires **PolicyDecision**, local access is read-first where it inspects state, and transitions are recorded through **RuntimeJournal**.

### 12.2 Policy And Action Gate



### 12.3 Policy Matrix

| Action class |       |           | Data class              |                       | Autonomy level |  | Authorization                          | Trace/Recovery                                                                      |
|--------------|-------|-----------|-------------------------|-----------------------|----------------|--|----------------------------------------|-------------------------------------------------------------------------------------|
| Read         | local | meta-data | Public sensitivity      | or low-local metadata | Observation    |  | None by default                        | Log access when used for reasoning.                                                 |
| Draft text   |       |           | Any allowed text bundle | con-                  | Draft creation |  | None unless sensitive scope is crossed | Retain as conversation or generated material, not authority or candidate by itself. |

| Action class                                                                     | Data class                            | Autonomy level          | Authorization                                                                   | Trace/Recovery                                                                                                          |
|----------------------------------------------------------------------------------|---------------------------------------|-------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Update known material                                                            | Scoped module material                | Bounded mutation        | Required provenance, metadata, scope, or sensitivity changes when truth changes | Versioned <b>KnowledgeRecord</b> with correction and rollback; <b>KnowledgeCandidate</b> if boundary change is pending. |
| Invoke internal action                                                           | Small task-scoped non-sensitive input | Bounded internal action | Policy allow required                                                           | Requested action, input class, result, policy outcome, and recovery expectation recorded.                               |
| Contact remote service                                                           | Minimized and classified transfer set | External action         | Explicit policy authorization                                                   | Transfer record, provider, purpose, and retention assumption recorded.                                                  |
| Modify credentials, identity, legal, financial, destructive, or executable state | Sensitive or critical data            | Critical action         | Strong human authorization                                                      | Block unless recovery or explicit irreversible authorization exists.                                                    |

Knowledge-boundary updates, routing mutation, workflow mutation, and learning lifecycle behavior follow the same authority rule: a pending knowledge-boundary change is a **KnowledgeCandidate**; an evaluated behavior improvement is a **LearningCandidate**; neither becomes active merely by being produced.

## 12.4 Learning Lifecycle

Learning follows a controlled lifecycle: observation, evaluation, **LearningCandidate** creation, evidence collection, evaluation, promotion, active monitoring, and rollback. Knowledge handling remains distinct: a stored **KnowledgeRecord** is not learning, and a pending change to its owning knowledge boundary is a **KnowledgeCandidate**. Routing preferences, workflow improvements, and inference-selection improvements remain inactive unless the appropriate learning promotion rule validates them for a defined scope.

# Threat Model

## 13.1 Threat Taxonomy

| Threat                                       | MADRE exposure                                                                                                                          | Primary control                                                                                                           | Verification surface                                           |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Source as instruction                        | Untrusted text attempts to alter policy, action execution, or knowledge handling by being interpreted as an instruction.                | Source classification and instruction/data separation.                                                                    | Adversarial context scenario.                                  |
| Sensitive disclosure                         | Private context leaks to a model, action, or remote service.                                                                            | Data classification, minimization, and remote-transfer authorization.                                                     | Disclosure review.                                             |
| Excessive agency                             | <code>InferenceOutput</code> causes over-broad action.                                                                                  | Autonomy levels and action-class policy.                                                                                  | Policy matrix review.                                          |
| Unsafe action execution                      | Action invocation produces side effects outside task scope.                                                                             | Policy-gated internal action boundary; remote, host-command, filesystem, and mutation actions require stronger authority. | Allowed-action and blocked-action scenarios.                   |
| Incorrect truth authority for known material | A <code>KnowledgeRecord</code> is used as factual authority despite unsuitable <code>truthLevel</code> or <code>truthAuthority</code> . | Required knowledge metadata, intended-use checks, correction, and reclassification path.                                  | Non-authoritative knowledge-use and truth-correction scenario. |
| Inspection misuse                            | Local console is treated as authority or mutation path.                                                                                 | Read-first inspection boundary and shared policy model.                                                                   | Local inspection negative scenario.                            |

| Threat                              | MADRE exposure                                                                                                                | Primary control                                                                   | Verification surface                         |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------|
| Supply-chain or provider dependency | Model, dependency, or service changes behavior or availability.                                                               | Replaceable <b>InferenceBackend</b> and dependency policy.                        | Inference substitution review.               |
| Remote transfer risk                | Local data is sent outside the user's domain without classification, minimization, authorization, or recorded trace.          | Local-first default and remote-transfer policy boundary.                          | Remote-contact blocked scenario.             |
| Unbounded consumption               | Delayed work consumes excessive compute, storage, or cost.                                                                    | Resource budgets, queue limits, and status inspection.                            | Resource benchmark.                          |
| Trace/recovery gaps                 | A result cannot be explained, corrected, invalidated, superseded, rolled back, or detached.                                   | <b>RuntimeJournal</b> , append-only <b>RuntimeEvent</b> , and recovery semantics. | Recovery trace scenario.                     |
| Inference output as false authority | Fluent <b>InferenceOutput</b> is treated as truth, policy, knowledge, behavior, routing, authority, or a candidate by itself. | Explicit generated-material handling and <b>PolicyDecision</b> boundary.          | Output non-authority/non-candidate scenario. |



# Quality Model

## 14.1 Quality Attribute Scenarios

| Attribute                     | Scenario                                                                                                                                  | Measure or evidence                                                                  |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Responsiveness</b>         | The foreground remains conversational while deeper work is queued.                                                                        | Foreground latency and clear work status.                                            |
| <b>Recoverability</b>         | A failed or unsafe delayed task can be inspected and corrected.                                                                           | <code>RuntimeJournal</code> trace, status transition, and roll-back/correction path. |
| <b>Auditability</b>           | A result can be explained through context, <code>InferenceRecord</code> , action, <code>PolicyDecision</code> , and verification records. | Complete journaled outcome evidence chain.                                           |
| <b>Maintainability</b>        | An <code>InferenceBackend</code> or action contract can be replaced without changing kernel authority.                                    | Contract and substitution review.                                                    |
| <b>Portability</b>            | A useful subset runs on ordinary local hardware.                                                                                          | Local profile and resource benchmark.                                                |
| <b>Security</b>               | Untrusted context cannot silently become instruction or policy.                                                                           | Prompt-injection and data-boundary scenarios.                                        |
| <b>Resource discipline</b>    | Delayed work is bounded by cost, time, storage, and compute policies.                                                                     | Queue and resource-use metrics.                                                      |
| <b>Privacy</b>                | Sensitive or unrelated local material is excluded from context and remote transfer unless scoped and policy-allowed.                      | Context exclusion and remote-transfer scenarios.                                     |
| <b>Model substitutability</b> | Boundary behavior is preserved across replaceable <code>InferenceProfile</code> and <code>InferenceBackend</code> choices.                | Inference substitution scenario.                                                     |
| <b>Context isolation</b>      | One task's context does not silently bleed into another topic, user, inference execution, action invocation, or candidate.                | Context scope and revocation review.                                                 |

| Attribute                                | Scenario                                                                                                                                                             | Measure or evidence                     |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <b>Explainability and inspectability</b> | Users and developers can see status, <code>PolicyDecision</code> , context evidence, <code>InferenceRecord</code> , action evidence, candidates, and recovery state. | Local read-first inspection scenario.   |
| <b>Testability</b>                       | Architecture claims can be checked through behavior scenarios and negative paths.                                                                                    | Requirements and decision trace review. |

# UX, Observability, And Recovery

## 15.1 Interaction Contract

| State                     | User-visible meaning                                                                                                                     |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Immediate answer          | The foreground has enough safe local context to answer.                                                                                  |
| Clarification needed      | Intent, scope, risk, or context is insufficient.                                                                                         |
| Queued reasoning          | Work has been recorded for delayed execution.                                                                                            |
| Running task              | Delayed reasoning or action work is active.                                                                                              |
| Blocked by policy         | A policy boundary prevents execution or promotion.                                                                                       |
| Authorization required    | The action crosses a configured authority boundary.                                                                                      |
| Failed task               | Work ended without acceptable result or recovery.                                                                                        |
| Completed task            | Result is available with <code>RuntimeJournal</code> trace evidence.                                                                     |
| Corrected or rolled back  | A previous result or knowledge-boundary change has been revised or reverted.                                                             |
| Invalidated or superseded | Prior evidence, source, or result is no longer active because it was corrected, revoked, reclassified, or replaced.                      |
| Deleted or detached       | Active generated material is removed from use where allowed while accountable <code>RuntimeEvent</code> history remains distinguishable. |
| Learning candidate        | An evaluated <code>LearningCandidate</code> proposal exists but is not active.                                                           |

## 15.2 Observability Surfaces

The local inspection surface must expose sessions, `ReasoningTask`, `RuntimeEvent`, context bundles, `InferenceRecord` and `InferenceOutput`, internal action evidence, `PolicyDecision`, failures, metrics, checkpoints, rollbacks, invalidations, supersessions, deleted/detached active material, `KnowledgeCandidate`, `LearningCandidate`, and recovery actions. The inspection surface is not a privileged bypass; it is a read-first view over the same policy-governed runtime.

# Evaluation And Benchmark Specification

**MADRE** defines an architectural evaluation surface for testing whether the runtime preserves local authority, governed context, bounded action use, evaluated learning promotion, journaled traceability, and recoverable delayed reasoning.

## 16.1 Benchmark Families

| Benchmark family                      | Purpose                                                                                                                                              |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Functional correctness</b>         | Confirm expected runtime behavior for foreground-lane continuity and delayed work.                                                                   |
| <b>Delayed reasoning quality</b>      | Evaluate evidence handling, decomposition, verification, and result delivery.                                                                        |
| <b>Foreground triage</b>              | Confirm unsupported evidence-heavy answers are clarified or delayed rather than fabricated.                                                          |
| <b>Context isolation</b>              | Confirm unrelated or sensitive knowledge does not enter the wrong bundle.                                                                            |
| <b>Inference substitution</b>         | Confirm replaceable <code>InferenceProfile</code> and <code>InferenceBackend</code> choices preserve the same boundary behavior and expose failures. |
| <b>Policy enforcement</b>             | Confirm action, data, autonomy, authorization, and recovery rules are enforced.                                                                      |
| <b>Bounded internal action</b>        | Confirm the bounded internal action is policy-gated, recorded, and harmless.                                                                         |
| <b>Blocked unsafe actions</b>         | Confirm remote, host-command, filesystem, mutation, and critical-action attempts do not execute under bounded internal-action authority.             |
| <b>Inference output non-authority</b> | Confirm <code>InferenceOutput</code> remains generated material, neither authority nor a candidate by itself.                                        |
| <b>Known-material truth control</b>   | Confirm a <code>KnowledgeRecord</code> with non-authoritative truth metadata is not used as factual truth.                                           |

| Benchmark family                    | Purpose                                                                                                                                                                 |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Learning promotion control</b>   | Confirm a <code>LearningCandidate</code> remains inactive until evaluated promotion and that saving <code>KnowledgeRecord</code> is not learning.                       |
| <b>Runtime journal traceability</b> | Confirm <code>ReasoningTask</code> , <code>RuntimeEvent</code> , <code>PolicyDecision</code> , and inference references expose lifecycle and recovery evidence locally. |
| <b>Read-first observability</b>     | Confirm CLI/status and local console inspection expose lifecycle, policy, context, inference, action, knowledge/learning transition, and recovery evidence.             |
| <b>Recovery</b>                     | Confirm failed or unsafe outputs can be inspected, corrected, deleted/detached, invalidated, superseded, or rolled back.                                                |
| <b>Resource use</b>                 | Measure latency, storage, queue time, model cost, and local compute use.                                                                                                |
| <b>Reproducibility</b>              | Preserve enough configuration, context, and event history to review outcomes.                                                                                           |
| <b>Hallucination resistance</b>     | Confirm <code>InferenceOutput</code> does not become truth, knowledge, action authority, routing, behavior, or candidate status by itself.                              |

# Governance, Deployment, And Limits

---

## 17.1 Secure Engineering Principles

**MADRE** requires secure defaults for local credentials, dependency selection, governed external boundaries, secrets handling, reviewable release records, and vulnerability response. Security evidence must cover both conventional software risks and model-mediated risks such as prompt injection, poisoned retrieval, unsafe output handling, excessive agency, and remote disclosure.

## 17.2 Open-Source Governance

**MADRE** is released under GPL-3.0. The open-source governance model must preserve license clarity, dependency review, contribution rules, security disclosure, release integrity, and community responsibility. Openness is part of the project value only when users and developers can inspect, adapt, and verify the runtime without surrendering private data to a platform owner.

## 17.3 Deployment And Operational Assumptions

The baseline deployment target is an ordinary local device with local storage, local configuration, optional local model execution, and optional remote integrations governed by policy. The operational profile includes storage paths, model setup assumptions, backup and restore, migration, diagnostics, log retention, offline behavior, and upgrade or rollback expectations.

Deployment assumptions remain conceptual at this level of specification. The product architecture defines authority, locality, inspection, recovery, and model-runtime and action boundaries without reducing deployment to a single operational shape.

## 17.4 Risks, Limitations, And Excluded Functions

**MADRE** limits active executable behavior by authority class. Workflow authoring, self-learning, speech interaction, graphical product surfaces, distributed inference, organization-scale deployment, and model fine-tuning remain governed by the same local-first, policy-governed, model-agnostic, auditable, and recoverable runtime boundary as the rest of the product.

## Requirements Catalogue

| ID             | Requirement summary                                                                                                                                                                                            | Decision trace       |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>FR-001</b>  | Preserve foreground-lane conversation while delaying work that requires evidence, verification, internal actions, or stronger context.                                                                         | D-001, D-004.        |
| <b>FR-002</b>  | Record delayed reasoning as <b>ReasoningTask</b> with <b>RuntimeEvent</b> lifecycle and recovery path through <b>RuntimeJournal</b> .                                                                          | D-001, D-003, D-008. |
| <b>FR-003</b>  | Construct context bundles from sourced, classified, minimized, scoped, provenanced, and revocable information.                                                                                                 | D-001, D-005, D-007. |
| <b>FR-004</b>  | Execute model-backed work through <b>ModelAction</b> , allowed <b>ModelBinding</b> , and replaceable <b>InferenceBackend</b> , producing <b>InferenceRecord</b> and non-authoritative <b>InferenceOutput</b> . | D-001, D-006.        |
| <b>FR-005</b>  | Expose executable behavior only as declared, bounded, traceable internal actions or explicitly authorized external actions.                                                                                    | D-007, D-009.        |
| <b>FR-006</b>  | Keep pending <b>KnowledgeCandidate</b> change separate from evaluated <b>LearningCandidate</b> improvement and require promotion for each.                                                                     | D-011.               |
| <b>FR-007</b>  | Expose local request entry and read-first inspection.                                                                                                                                                          | D-010.               |
| <b>FR-008</b>  | Represent recovery outcomes explicitly.                                                                                                                                                                        | D-003, D-008.        |
| <b>SEC-001</b> | Treat prompt/source material as data rather than instruction; treat <b>InferenceOutput</b> as generated material rather than authority or candidate by itself.                                                 | D-004, D-005, D-007. |
| <b>OPS-001</b> | Keep local access surfaces governed by runtime authority.                                                                                                                                                      | D-010.               |

## Traceability Matrix

| Requirement   | Concern                         | Architecture element                                                                               | Evidence family                                       |
|---------------|---------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>FR-001</b> | Responsiveness and user control | Complex Agent foreground lane and delayed reasoning split                                          | Interaction and latency scenarios.                    |
| <b>FR-002</b> | Recoverability and traceability | <b>ReasoningTask</b> , <b>RuntimeEvent</b> , and <b>RuntimeJournal</b>                             | Recovery and status scenarios.                        |
| <b>FR-003</b> | Privacy and correctness         | Governed context bundle                                                                            | Context isolation scenarios.                          |
| <b>FR-004</b> | Provider independence           | <b>ModelAction</b> , <b>InferenceBackend</b> , <b>InferenceRecord</b> , and <b>InferenceOutput</b> | Inference substitution and output-boundary scenarios. |
| <b>FR-005</b> | Action safety                   | Internal Action Registry, action contract, and external action policy                              | Allowed-action and blocked-action scenarios.          |
| <b>FR-006</b> | Controlled evolution            | Separate knowledge-change and learning-promotion candidates                                        | Knowledge-boundary and learning-promotion review.     |
| <b>FR-007</b> | Local operability               | CLI plus read-first local console                                                                  | Local access and inspection scenario.                 |
| <b>FR-008</b> | Recoverability                  | <b>RuntimeEvent</b> history and recovery semantics                                                 | Recovery trace.                                       |



# Threat Catalogue

| Threat family                                      | Primary boundary                        | Initial response                                                                                                            |
|----------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Source as instruction                              | Context bundle construction             | Separate source material from instruction and policy.                                                                       |
| Sensitive disclosure                               | Data classification and remote transfer | Minimize, redact, authorize, and record disclosures.                                                                        |
| Unsafe action execution                            | Action and workflow contracts           | Require action-class policy, bounded internal-action scope, and recovery expectation.                                       |
| Known material used under incorrect truth metadata | KnowledgeRecord use                     | Enforce provenance, <code>truthLevel</code> , <code>truthAuthority</code> , intended use, correction, and reclassification. |
| Inspection bypass                                  | Local console and status surfaces       | Read-first inspection with no privileged mutation path.                                                                     |
| Inference-output false authority                   | <code>InferenceOutput</code> boundary   | Treat <code>InferenceOutput</code> as generated material, not authority or candidate by itself.                             |
| Provider dependency                                | Local Model Runtime Boundary            | <code>InferenceBackend</code> substitution and local-first default.                                                         |
| Unbounded consumption                              | Scheduler and resource controls         | Queue, compute, storage, and cost limits.                                                                                   |

## Benchmark Catalogue

| Benchmark                    | Observed property                                                                                                             | Evidence                                                           |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Foreground response          | Conversational continuity under delayed work                                                                                  | Latency and interaction-state record.                              |
| Delayed work durability      | <b>ReasoningTask</b> survives interruption and exposes status                                                                 | <b>RuntimeEvent</b> lifecycle in <b>RuntimeJournal</b> .           |
| Context isolation            | Only scoped material with suitable metadata enters a task context                                                             | Context bundle and <b>KnowledgeRecord</b> inspection.              |
| Policy enforcement           | Disallowed action remains blocked or requires authorization                                                                   | Policy decision record.                                            |
| Bounded internal action      | Bounded internal action remains local, policy-gated, and traceable                                                            | Action trace.                                                      |
| Inference output boundary    | <b>InferenceOutput</b> remains generated material, not authority or candidate by itself                                       | <b>InferenceRecord</b> , output-handling, and negative-path trace. |
| Knowledge truth authority    | Non-authoritative <b>KnowledgeRecord</b> is not used as factual truth                                                         | Truth-metadata use and correction trace.                           |
| Learning promotion control   | <b>LearningCandidate</b> remains inactive and storing <b>KnowledgeRecord</b> is not learning                                  | Promotion and negative-path trace.                                 |
| Runtime journal traceability | Local trace correlates work, events, policies, inference, and recovery                                                        | <b>RuntimeJournal</b> inspection trace.                            |
| Read-first inspection        | Local console exposes status, trace, policy, context, inference, action, knowledge/learning transition, and recovery evidence | Inspection trace.                                                  |
| Recovery                     | Failed or unsafe result can be corrected or rolled back                                                                       | Recovery trace.                                                    |
| Inference substitution       | <b>InferenceBackend</b> replacement preserves the boundary contract                                                           | Inference boundary comparison.                                     |

# Runtime Domain Model

**MADRE** principles are rooted in the domain architecture rather than brute computation. It is designed as a reusable and reliable framework for building agentic systems.

This chapter defines the runtime class map for **MADRE** at architecture level. It describes the minimum set of classes currently needed to express module ownership, agent execution, delayed reasoning, governed context, bounded action execution, produced artifacts, policy outcomes, event trace, and storage access without turning unknown implementation mechanics into false domain classes.

## 22.1 Design Rule

A class exists in this map only when it has stable runtime identity, stable ownership boundary, stable lifecycle, stable responsibility that cannot be expressed as a value inside another class, or required participation in the canonical **MADRE** flow.

State transitions, boundary composition, quality gates, rollback rules, action classification, and storage schema are open runtime design areas. The current class surface names only the stable runtime objects required by the canonical flow.

## 22.2 Minimal Canonical Flow

1. An access-surface adapter made available by the kernel delivers interaction to its owning module
2. The Core Module is the default module for unknown, empty, or invalid targets
3. A module creates a **ReasoningRequest** for work that may remain local or be routed to another module
4. The kernel routes **ReasoningRequest** objects, records events, and schedules plans; it does not reason, inspect module knowledge, or evaluate boundaries
5. **ReasoningModule.handle(request)** returns a **ReasoningResponse**: answer, clarification, block, delegation, plan creation, or failure
6. Optional **Workflow** blueprints help structure delayed work into **ReasoningTask** occurrences
7. Module orchestration creates **AgentRequest** objects for selected **MADREAgent** and **MADREAction** execution
8. The agent/action path returns an **AgentResponse** containing produced **ReasoningArtifact** objects, block material, or failure material

9. Runtime history is appended as `RuntimeEvent` entries in `RuntimeJournal`; process reconstruction belongs to event and storage queries

## 22.3 Conventions

### 22.3.1 Type Conventions

- **UUID:** Stable identifier
- **Text:** Human-readable or structured textual description
- **Instant:** Timestamp
- **Ref<T>:** Stable reference to another runtime object
- **List<T>:** Ordered collection
- **Set<T>:** Unordered collection without duplicates
- **Map<K,V>:** Key-value structure
- **Boolean:** True or false

### 22.3.2 Process Conventions

- **minimizeContext:** context adapted to the target scope, with sensitive or unrelated material omitted
- **boundary composition:** the construction of the execution contract from scope, sensitivity, risk, roll-back, policy, and involved actors
- **state transition:** guarded process, not a plain enum assignment

### 22.3.3 Class Inventory

| Included classes                                              | Open runtime areas                                                                 |
|---------------------------------------------------------------|------------------------------------------------------------------------------------|
| MADREKernel, DBAccessor, RuntimeJournal, RuntimeEvent         | Concrete storage schema, scheduler implementation, access-surface implementations. |
| ReasoningModule, ReasoningRequest, ReasoningResponse          | Module-selection heuristics, request-state internals.                              |
| ReasoningPlan, ReasoningTask, Workflow                        | Plan-state internals, task-state internals, quality-gate internals.                |
| MADREAgent, SimpleAgent, TwinAgent, ComplexAgent, SystemAgent | Threading model and concrete lane scheduling.                                      |

| Included classes                                 | Open runtime areas                                                                |
|--------------------------------------------------|-----------------------------------------------------------------------------------|
| AgentRequest, AgentResponse, MADREAction         | Action classification enums, model-specific adapter classes.                      |
| ContextBundle, PolicyDecision, BoundaryContract, | Boundary-part taxonomy and policy-rule engine internals.                          |
| ReasoningArtifact, KnowledgeRecord               | Artifact subtype catalogue, knowledge-candidate and learning-candidate lifecycle. |

### 22.3.4 Structural Relations

- **Kernel:**

MADREKernel registers ReasoningModule objects.

MADREKernel assigns one registered module as the Core Module role.

Core Module is a role, not a class.

MADREKernel routes ReasoningRequest objects, assigns DB access, records events, and schedules plans.

- **Module:**

ReasoningModule owns bounded data, knowledge, context governance, allowed agents, allowed actions, and workflows.

ReasoningModule creates and receives ReasoningRequest objects.

ReasoningModule may answer directly, block, clarify, delegate, or create a ReasoningPlan.

- **Agent and action:**

MADREAgent is abstract; concrete agents execute AgentRequest objects.

SimpleAgent, TwinAgent, ComplexAgent, and SystemAgent are subclasses of MADREAgent.

MADREAction is executable behavior available to an agent.

Inference is represented as action execution that produces response material in this runtime map.

- **Planning:**

Workflow is a reusable blueprint, not an action, actor, or scheduled object.

ReasoningRequest is routed.

ReasoningPlan is scheduled.

ReasoningTask is a passive planned occurrence inside exactly one ReasoningPlan.

- **Artifacts and trace:**

ReasoningArtifact is produced material from reasoning or action execution.

KnowledgeRecord is module-owned known material and is not truth by default.

PolicyDecision records a boundary or policy evaluation result.

RuntimeJournal appends RuntimeEvent history.

RuntimeEvent records what happened but does not authorize what happened.

## 22.4 Class Definitions

### 22.4.1 MADREKernel

- **Responsibilities:**

- Register reasoning modules.
- Assign the Core Module role.
- Schedule `ReasoningPlan` objects.
- Route `ReasoningRequest` objects.
- Own low-level runtime bindings for storage, scheduling, access surfaces, device resources, and host services.
- Assign `DBAccessor` objects to modules.
- Append runtime events through `RuntimeJournal`.

- **Boundaries:**

- The kernel does not reason.
- It does not plan module work.
- It does not execute actions.
- It does not invoke models directly.
- It does not evaluate boundary compatibility.
- It does not own module knowledge.
- It does not promote knowledge or learn.
- It does not decide semantic module suitability.

## MADRE Agentic System

Class Definition  
**MADREKernel**

**MADREKernel**: MADREKernel provides request routing, storage assignment, scheduling, and trace.

**Attributes:**

- kernelId: UUID
- coreModule: Ref<ReasoningModule>
- moduleRegistry: Map<Ref<ReasoningModule>, ReasoningModule>
- dbAccessors: Map<Ref<ReasoningModule>, DBAccessor>
- runtimeJournal: RuntimeJournal

**Operations:**

- + route(request: ReasoningRequest): ReasoningResponse
- + schedule(plan: ReasoningPlan): RuntimeEvent
- + record(event: RuntimeEvent): void
- registerModule(module: ReasoningModule): RuntimeEvent
- assignCoreModule(moduleRef: Ref<ReasoningModule>): RuntimeEvent
- assignDBAccessor(moduleRef: Ref<ReasoningModule>, accessor: DBAccessor): RuntimeEvent

### Notes:

Runtime provider for request routing, storage access, event recording, and scheduled-plan activation.

Flow: an access surface or module submits a **ReasoningRequest** to **route**; empty, unknown, or invalid targets resolve to the Core Module by routing rule.

Flow: module calls **schedule** after creating a **ReasoningPlan**; the kernel schedules the plan but does not execute it.

Flow: modules and runtime adapters call **record** to preserve **RuntimeEvent** evidence through the runtime journal.

Access-surface adapter transport remains outside this runtime class surface.

Boundary and policy evaluation belong to the module, agent, action, and boundary contract, not to the kernel.

## 22.4.2 ReasoningModule

### • Responsibilities:

- Own a bounded data and knowledge scope.
- Own module-local context governance.
- Own allowed agents, actions, and workflows.
- Create and receive **ReasoningRequest** objects.
- Answer directly when safe and sufficient.
- Create **ReasoningPlan** objects when structured or delayed work is needed.
- Advance scheduled plans owned by the module.
- Create minimized delegated **ReasoningRequest** objects when another module is required.
- Ensure output is minimized to consumer scope.

### • Boundaries:

A module does not become the kernel.

It does not own knowledge outside its boundary.

It does not directly access another module's private data.

It does not bypass policy, context governance, boundary-contract construction, action risk, or journal trace.

## MADRE Agentic System

Class Definition  
ReasoningModule

**ReasoningModule:** ReasoningModule owns module-local behavior and knowledge boundaries.

**Attributes:**

- moduleId: UUID
- moduleManager: Ref<SystemAgent>
- agentCatalog: List<Ref<MADREAgent>>
- workflows: List<Workflow>
- dbAccessor: DBAccessor

**Operations:**

- + handle(request:ReasoningRequest): ReasoningResponse
- + advance(plan:ReasoningPlan): RuntimeEvent
- createPlan(request:ReasoningRequest): ReasoningPlan
- createDelegatedRequest(request:ReasoningRequest): ReasoningRequest
- buildAgentRequest(task:ReasoningTask): AgentRequest
- composeBoundary(request:AgentRequest): BoundaryContract

### Notes:

Bounded reasoning and data-governance unit for a topic, domain, or operational scope.

Relationships: owns or is allowed to use module agents, actions, workflows, and knowledge records.

Flow: MADREKernel.route(request) calls handle on the selected module after routing has resolved the receiving module.

Flow: handle may return direct response, clarification, block, delegated request, plan-created response, or failure response.

Flow: kernel scheduling calls advance on the owning module for a scheduled ReasoningPlan; the module then creates AgentRequest objects and invokes selected agents.

The module constructs the boundary contract for agent/action execution from request, context, policy, action, rollback, and actor material.

### 22.4.3 MADREAgent

- **Responsibilities:**

Execute concrete AgentRequest objects.

Use the explicit request context plus its own module-bound execution context.

Invoke owned MADREAction objects only through a valid AgentRequest.

Return artifacts, block material, or failure material inside AgentResponse.

Preserve no module-private knowledge outside the owning module boundary.



- **Boundaries:**

An agent does not own kernel resources.

It does not widen its own authority.

It does not access module knowledge except through provided context.

It does not promote generated material into knowledge or learning.

## MADRE Agentic System

Class Definition  
MADREAgent

*<AbstractClass>*

**MADREAgent**: MADREAgent executes module-authorized requests through declared actions.

**Attributes:** - agentId: UUID

**Operations:** + execute(request:AgentRequest): AgentResponse

### Notes:

Runtime actor used by a module to coordinate or execute **AgentRequest** objects.

Caller: module-owned orchestration invokes the agent after it has built the **AgentRequest** and boundary contract.

Relationships: owns or is assigned allowed actions by module-owned behavior.

Concrete subclasses define execution style; no profile enum is part of the runtime object.

### 22.4.4 SimpleAgent

- **Responsibilities:**

Execute a direct agent request through its selected action.

Return one **AgentResponse**.

Avoid learning and review-lane behavior.

- **Boundaries:**

A simple agent does not double-check itself.

It does not reflect or learn.

It does not gain authority beyond its module and request contract.

## MADRE Agentic System

Class Definition  
SimpleAgent

**SimpleAgent**: SimpleAgent directly executes module-authorized requests.

**Attributes:**

**Operations:** -NoValue-

### Notes:

Concrete **MADREAgent** for direct execution where no twin review or reflection behavior is required.

Inheritance: subclass of **MADREAgent**.

## 22.4.5 TwinAgent

- **Responsibilities:**

- Produce an initial response path.
- Review or double-check the produced response before dispatch.
- Return a checked **AgentResponse**.

- **Boundaries:**

- A twin agent does not learn.
- Its review path does not widen action authority.
- It cannot dispatch unchecked material as final response when review is required.

### MADRE Agentic System

Class Definition  
TwinAgent

**TwinAgent**: TwinAgent executes and checks a response before dispatch.

**Attributes:**

**Operations:** -NoValue-

**Notes:**

Concrete **MADREAgent** for response plus review behavior.

Inheritance: subclass of **MADREAgent**.

Review remains internal agent behavior.

## 22.4.6 ComplexAgent

- **Responsibilities:**

- Coordinate foreground response behavior.
- Coordinate deeper reasoning behavior when a request requires further work.
- Coordinate reflection behavior over previous responses, outcomes, and feedback.

- **Boundaries:**

- A complex agent does not gain extra authority from having more lanes.
- Reflection may propose improvement but cannot activate learning by itself.
- Foreground response must remain inside the module and request boundary contract.

## MADRE Agentic System

Class Definition  
ComplexAgent

**ComplexAgent**: ComplexAgent coordinates foreground, reasoning, and reflection behavior.

Attributes:

Operations: -NoValue-

Notes:

Concrete MADREAgent for foreground, reasoning, and reflection coordination.

Inheritance: subclass of MADREAgent.

Lane scheduling and threading belong to concrete execution design.

### 22.4.7 SystemAgent

- **Responsibilities:**

Bind a complex agent to one ReasoningModule.

Manage module-scoped context and module-scoped learning behavior.

Serve as module manager when assigned that role.

- **Boundaries:**

A system agent never leaves its assigned module.

It does not become the kernel.

It does not manage another module's private knowledge.

## MADRE Agentic System

Class Definition  
SystemAgent

**SystemAgent**: SystemAgent is a module-bound complex agent for module management behavior.

Attributes:

Operations: -NoValue-

Notes:

Module-bound ComplexAgent. It may hold the module manager role.

Module binding and manager-role assignment are owned by the module, not by an inverse module reference.

MainAgent remains a role assigned to a module-owned agent, not a separate class.

### 22.4.8 ReasoningRequest

- **Responsibilities:**

Preserve the reasoning objective.

Preserve origin context and routing intent.

- Preserve minimized originator context references.
- Preserve boundary-relevant request material.
- Preserve request-chain information sufficient to avoid loops.
- Carry the expected result shape at request level.

- **Boundaries:**

- A request does not build plans.
- It does not schedule itself.
- It does not execute actions.
- It does not select target modules by itself.
- It does not minimize itself.
- It does not reveal originating plan internals.

## MADRE Agentic System

### Class Definition ReasoningRequest

**ReasoningRequest :** ReasoningRequest records a reasoning need and routing boundary.

- requestId: UUID

**Attributes:** - requestChain: List<UUID>  
 - objective: Text

**Operations:** -NoValue-

### Notes:

Passive module-level request routed by the kernel.

Relationships: may carry minimized origin context and routing evidence, but it does not require a caller to know a concrete module reference.

Invalid, empty, or unavailable targets resolve to the Core Module through `MADREKernel.route(request)`.

Request state and target selection are produced by routing and journaled runtime transitions.

## 22.4.9 ReasoningResponse

- **Responsibilities:**

- Preserve the result of module-level request handling.
- Represent answer, clarification, block, delegation, plan creation, or failure.
- Carry response material without making it knowledge, policy, or authority.

- **Boundaries:**

- A reasoning response does not execute actions.
- It does not schedule itself.
- It does not promote contained material into knowledge or learning.

## MADRE Agentic System

### Class Definition ReasoningResponse

**ReasoningResponse**: ReasoningResponse records module-level handling output.

**Attributes:**

- responseId: UUID
- producedArtifacts: List<ReasoningArtifact>
- delegatedRequest: Ref<ReasoningRequest>
- createdPlan: Ref<ReasoningPlan>

**Operations:** -NoValue-

### Notes:

Module-level response envelope.

The handled request is known by the call and journaled event; the response does not own an inverse origin reference.

Response categories and status transitions belong to runtime handling.

## 22.4.10 ReasoningPlan

### • Responsibilities:

Represent the selected reasoning strategy.

Own task occurrences.

Preserve delegated request links.

Preserve current plan state, results, failures, and produced artifacts.

Support scheduling by the kernel.

Support inspection through journaled events.

### • Boundaries:

A plan does not execute itself.

It does not allocate kernel resources.

It does not bypass module ownership.

It does not expose another module's internals.

It does not promote results into knowledge or learning.

## MADRE Agentic System

Class Definition  
ReasoningPlan

**ReasoningPlan**: ReasoningPlan owns delayed reasoning task structure and result links.

**Attributes:** - planId: UUID

**Operations:** -NoValue-

### Notes:

Schedulable delayed-work object created from a **ReasoningRequest**.

Relationships: links to its source request, owning module, task occurrences, applied workflows, and produced result artifacts.

Plan status is a guarded runtime projection reconstructed from **RuntimeEvent**.

### 22.4.11 ReasoningTask

- **Responsibilities:**

Bind a concrete planned step to a responsible module, an agent, and an **AgentRequest**.

Preserve dependency information.

Preserve expected result.

Preserve current status, result reference, failure reference, or delegated request reference.

- **Boundaries:**

A task does not execute itself.

It does not transition itself independently.

It does not create delegated requests.

It does not inspect another module's plan internals.

It does not own scheduling.

It does not promote results into knowledge or learning.

## MADRE Agentic System

Class Definition  
ReasoningTask

**ReasoningTask:** ReasoningTask is a passive planned execution record inside a ReasoningPlan.

- taskId: UUID
- dependencies: List<Ref<ReasoningTask> >

**Attributes:** - assignedAgent: Ref<MADREAgent>

- agentRequest: AgentRequest
- boundaryContract: BoundaryContract

**Operations:** -NoValue-

### Notes:

Passive planned occurrence inside exactly one ReasoningPlan.

The owning plan contains the task occurrence; the task does not own an inverse plan reference.

Relationships: binds responsible module context, selected agent, concrete agent request, boundary contract, and dependencies.

Status, results, failures, quality gates, and delegated request links are guarded runtime projections.

Task dispatch is performed by module orchestration, not by the task itself.

### 22.4.12 AgentRequest

- **Responsibilities:**

Preserve the concrete invocation objective.

Preserve caller, target agent, target action, parameters, context, and composed boundary contract.

Carry only the minimized context needed for execution.

- **Boundaries:**

An agent request does not replace a reasoning request.

It does not schedule work.

It does not select its own authority.

It does not make context valid by containing it.

## MADRE Agentic System

Class Definition  
AgentRequest

**AgentRequest**: AgentRequest carries the concrete invocation boundary.

**Attributes:**

- agentRequestId: UUID
- parameters: Map<Text,Text>
- prompt: Text
- context: ContextBundle
- moduleScope: Ref<ReasoningModule>
- targetAgent: Ref<MADREAgent>
- targetAction: Ref<MADREAction>
- boundaryContract: BoundaryContract

### Notes:

Passive action-level invocation object sent to an agent or action.

Relationships: links caller, target agent, target action, and context bundle.

The request does not prove its own validity; it carries the already-composed execution contract.

### 22.4.13 AgentResponse

- **Responsibilities:**

Preserve the result of agent or action execution.

Wrap produced **ReasoningArtifact** objects and raw generated material when applicable.

Represent successful output, block material, refusal material, or failure material.

- **Boundaries:**

An agent response is not knowledge by default.

It is not policy.

It is not authority.

It does not activate learning or route itself.



## MADRE Agentic System

Class Definition  
AgentResponse

**AgentResponse:** AgentResponse wraps agent/action execution output.

**Attributes:**

- agentResponseId: UUID
- artifacts: List<ReasoningArtifact>

**Operations:** -NoValue-

### Notes:

Concrete response envelope returned by a MADREAgent or MADREAction.

The invoking AgentRequest is known by the call and journaled event; the response does not own an inverse origin reference.

Raw inference output, action output, files, patches, reports, and similar material are represented through produced artifacts inside this response.

Response status, block, refusal, and failure details are response material and runtime events.

### 22.4.14 MADREAction

- **Responsibilities:**

Declare minimal executable purpose.

Execute only through a concrete AgentRequest.

Return an AgentResponse representing result, block, refusal, or failure.

- **Boundaries:**

An action does not own scheduling.

It does not grant authority to itself.

It does not widen its owning agent's authority.

It does not make generated material true or known.

## MADRE Agentic System

Class Definition  
MADREAction

**MADREAction:** MADREAction executes bounded behavior and returns artifacts through AgentResponse.

**Attributes:** - actionId: UUID

**Operations:** + run(request:AgentRequest): AgentResponse

### Notes:

Executable capability available to an agent under a concrete AgentRequest.

Caller: the selected MADREAgent; direct module, kernel, or model invocation is outside the action execution path.

Relationships: executed through an owning agent and constrained by the invoking module scope.

Action classification, risk, rollback, and policy detail are part of the boundary contract and action declaration mechanics.

### 22.4.15 Workflow

- **Responsibilities:**

- Preserve reusable task structure.
- Describe task templates and dependency shape.
- Describe expected artifacts or validation gates.
- Be applied by authorized module-owned behavior.
- Never execute directly.

- **Boundaries:**

- A workflow is not a plan.
- It is not a scheduler.
- It is not an actor.
- It is not an action.
- It does not allocate resources, invoke agents, execute actions, or create delegated requests.

#### MADRE Agentic System

Class Definition  
Workflow

**Workflow:** Workflow is reusable structure, not executable runtime work.

**Attributes:** - workflowId: UUID

#### Notes:

- Reusable blueprint for structuring a **ReasoningPlan**.
- Authorized module-owned behavior applies workflows to create or refine task occurrences.

### 22.4.16 ContextBundle

- **Responsibilities:**

- Preserve selected context references.
- Preserve purpose and boundary.
- Ensure context can be checked against the target use.
- Prevent unrelated or over-privileged context exposure.

- **Boundaries:**

- A context bundle is not instruction.
- It is not truth.
- It is not policy.
- It is not authority.
- Included material retains its own provenance and trust limitations.

## MADRE Agentic System

Class Definition  
ContextBundle

**ContextBundle**: ContextBundle references governed context without making it authoritative.

**Attributes:** - bundleId: UUID

### Notes:

Minimized governed context prepared for one request, task, agent invocation, action, or inference call.

Relationships: created for a concrete use; selects artifact and knowledge material by reference without requiring the bundle itself to own module routing.

### 22.4.17 ReasoningArtifact

- **Responsibilities:**

Preserve produced material.

Preserve material needed by a response, task, or knowledge transition.

Represent generated output without making it true, known, learned, or authoritative.

- **Boundaries:**

An artifact is not knowledge by default.

It is not truth by default.

It is not learning by default.

It is not policy.

It is not authority.

It is not executable behavior by default.

## MADRE Agentic System

Class Definition  
ReasoningArtifact

<AbstractClass>

**ReasoningArtifact**: ReasoningArtifact preserves material without granting truth or authority.

**Attributes:** - artifactId: UUID

**Operations:** -NoValue-

### Notes:

Abstract produced material from reasoning or action execution.

Concrete artifact subtype taxonomy remains open.

AgentResponse wraps produced artifacts; the artifact does not own an inverse response or task origin reference.

Artifact material storage belongs to storage and event reconstruction design.

## 22.4.18 KnowledgeRecord

- **Responsibilities:**

Preserve material accepted into a module's knowledge boundary.

Preserve provenance.

Preserve intended use.

Preserve sensitivity and authority limits.

Preserve correction path.

Avoid treating stored material as truth by default.

- **Boundaries:**

A knowledge record is not factual truth by default.

Saving known material is not learning.

Updates, corrections, reclassification, deletion, detachment, or promotion require explicit module authority and journaled events.

## MADRE Agentic System

### Class Definition KnowledgeRecord

**KnowledgeRecord:** KnowledgeRecord is module-owned known material, not automatic truth.

**Attributes:**

**Notes:**

Module-owned known material with stronger ownership and truth-governance metadata than a generic artifact.

Relationships: owned by exactly one reasoning module boundary.

No separate knowledge identity is introduced here; it remains a **ReasoningArtifact** accepted into a module knowledge boundary.

Provenance, intended use, truth metadata, correction path, and lifecycle belong to the knowledge-boundary contract.

## 22.4.19 BoundaryContract

- **Responsibilities:**

Represent the composed execution boundary for one action path.

Combine module scope, agent, action, context, risk, rollback, policy, and involved actors through programmatic matching.

Prevent unmatchable execution parts from joining into a valid **AgentRequest**.

- **Boundaries:**

A boundary contract is not a generic profile.

It is not a pile of descriptive text attributes.

It does not make policy decisions by itself unless backed by the boundary mechanism.

## MADRE Agentic System

Class Definition  
BoundaryContract

**BoundaryContract**: BoundaryContract is the composed execution boundary for one action path.

**Attributes:** - contractId: UUID

**Operations:** -NoValue-

### Notes:

Programmatic contract carried by **AgentRequest**, **ReasoningTask**, and **AgentResponse**.

Boundary pieces remain part of boundary-composition design.

Invalid composition fails before action execution receives the request.

### 22.4.20 PolicyDecision

- **Responsibilities:**

Preserve allow, block, or authorization-required outcome.

Preserve the reason for the decision.

Link the decision to the affected request, action, context, or artifact boundary.

Provide audit evidence for later inspection.

- **Boundaries:**

A policy decision records authority outcome.

Generated material can inform it.

Generated material cannot create, override, or widen it by itself.

## MADRE Agentic System

Class Definition  
PolicyDecision

**PolicyDecision**: PolicyDecision records a policy outcome; it is not generated authority.

**Attributes:** - decisionId: UUID

### Notes:

Recorded result of checking a concrete boundary-sensitive operation.

Relationships: applies to a request, agent request, action, context bundle, artifact, or boundary contract.

Outcome shape, authorization-required mechanics, and reason structure remain part of the policy mechanism until stable.

## 22.4.21 RuntimeJournal

- **Responsibilities:**

Append runtime events.

Preserve traceability for requests, plans, tasks, agents, actions, artifacts, context, policy decisions, delegation, failures, and recovery.

Support inspection and reconstruction of read-only projections.

- **Boundaries:**

The journal records history.

It does not authorize.

It does not mutate.

It does not promote, correct, rollback, delete, or detach anything by itself.

### MADRE Agentic System

Class Definition  
RuntimeJournal

**RuntimeJournal**: RuntimeJournal preserves append-only runtime history.

**Attributes:**

- journalId: UUID
- dbAccessor: DBAccessor

---

**Operations:**

- + append(event:RuntimeEvent): void
- + trace(subjectId:UUID): List<RuntimeEvent>

### Notes:

Append-only local trace mechanism.

Relationships: uses DBAccessor to append and query event history.

Flow: MADREKernel.record(event) calls append.

Flow: local inspection calls trace to reconstruct read-only projections for a request, plan, task, agent request, action, artifact, or recovery path.

Trace projections are reconstructed by event and storage queries; domain objects do not duplicate database timestamps or schema fields.

## 22.4.22 DBAccessor

- **Responsibilities:**

Provide controlled storage access assigned by the kernel.

Store module content, knowledge, artifacts, and runtime events according to assigned scope.

Support event reconstruction without leaking another module's storage.

- **Boundaries:**

A DB accessor is not reasoning.

It is not policy.

It does not grant module authority by itself.

## MADRE Agentic System

Class Definition  
DBAccessor

**DBAccessor**: DBAccessor provides scoped storage access.

**Attributes:**

- accessorId: UUID
- owner: Ref<ReasoningModule>

**Operations:** -NoValue-

### Notes:

Runtime storage port owned and assigned by MADREKernel.  
 The concrete database schema is storage configuration.

### 22.4.23 RuntimeEvent

- **Responsibilities:**

- Preserve what happened.
- Preserve when it happened.
- Preserve who or what caused it.
- Preserve what object was affected.
- Preserve optional payload reference.

- **Boundaries:**

- An event records a transition, decision, result, or failure.
- It does not perform the transition it records.
- It does not authorize the transition it records.

## MADRE Agentic System

Class Definition  
RuntimeEvent

**RuntimeEvent**: RuntimeEvent records runtime history without authorizing it.

**Attributes:** - eventId: UUID

### Notes:

Append-only record of something that happened or was decided in the runtime.  
 Relationships: storage-backed event data links the affected subject, causal actor, event kind, and optional payload.  
 Timestamps, ordering, indexes, and reconstruction fields belong to the DB/event store configuration.

## 22.5 Validation Checklist

1. ReasoningRequest is routed, not scheduled
2. ReasoningPlan is scheduled, not directly executed by the kernel
3. ReasoningTask is passive and belongs to exactly one plan
4. ReasoningTask binds module, agent, concrete AgentRequest, and BoundaryContract
5. Workflow is a blueprint, not an executable action
6. MADREAction executes only under AgentRequest
7. MADREAgent cannot widen its own authority
8. BoundaryContract is composed before action execution
9. PolicyDecision records an authority outcome but is not generated by model output alone
10. ReasoningArtifact is not truth, knowledge, learning, policy, or authority by default
11. KnowledgeRecord is module-owned and is not factual truth by default
12. Kernel routing, recording, and scheduling do not become semantic reasoning
13. Intermodule delegation uses minimized ReasoningRequest, not source plan internals
14. Journal records history but does not perform transitions
15. Learning remains deferred until class boundaries and policy mechanics are stable.



## Glossary

| Term                    | Definition                                                                                                                                                                                                                 |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Internal action</b>  | A typed runtime-defined executable function available to modules through the Internal Action Registry with declared inputs, outputs, side-effect class, policy requirements, trace obligations, and recovery expectations. |
| <b>Core Module</b>      | A <b>ReasoningModule</b> role assigned by kernel registration for ordinary interaction first routing; it is not a module type.                                                                                             |
| <b>MainAgent</b>        | The role held by one module-owned <b>ComplexAgent</b> for ordinary module coordination; other <b>ComplexAgent</b> objects may exist without holding it.                                                                    |
| <b>ComplexAgent</b>     | A specialization class of <b>MADREAgent</b> capable to coordinate foreground answer, background reasoning, and reflection lanes without gaining additional authority.                                                      |
| <b>ModelAction</b>      | An <b>AgentAction</b> that uses governed context and an allowed <b>ModelBinding</b> to execute inference through <b>InferenceBackend</b> .                                                                                 |
| <b>ModelBinding</b>     | Allowed model/inference configuration assigned to a module-owned agent or model-backed action.                                                                                                                             |
| <b>InferenceProfile</b> | Configuration recipe for permitted inference execution; it is neither the model nor authority.                                                                                                                             |
| <b>InferenceBackend</b> | Replaceable execution boundary for local or explicitly authorized inference software.                                                                                                                                      |
| <b>InferenceRecord</b>  | The passive record of one inference execution and its configuration, backend, status, timing, and output reference.                                                                                                        |

| Term                                | Definition                                                                                                                                                                                                            |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>InferenceOutput</b>              | Generated material from an <b>InferenceRecord</b> ; it is not truth, policy, memory, knowledge, behavior, routing, authority, or candidate by itself.                                                                 |
| <b>Local Model Runtime Boundary</b> | Product-level phrasing for controlled local or authorized inference execution through <b>ModelBinding</b> , <b>InferenceProfile</b> , <b>InferenceBackend</b> , <b>InferenceRecord</b> , and <b>InferenceOutput</b> . |
| <b>ReasoningTask</b>                | Durable delayed-work lifecycle record linked to plan, module, request, runtime events, and recovery references.                                                                                                       |
| <b>RuntimeEvent</b>                 | Append-only record of runtime activity, including policy, action, inference, correction, and recovery transitions.                                                                                                    |
| <b>RuntimeJournal</b>               | Local runtime trace mechanism that records <b>ReasoningTask</b> , appends <b>RuntimeEvent</b> , and stores or references related payloads.                                                                            |
| <b>PolicyDecision</b>               | Explicit allow, block, or authorization-required outcome for an affected authority boundary.                                                                                                                          |
| <b>KnowledgeRecord</b>              | Module-owned known material with provenance, scope, sensitivity, intended use, truth metadata, correction, and lifecycle semantics; it is not factual truth by default.                                               |
| <b>KnowledgeCandidate</b>           | A pending change to a module knowledge boundary, not ordinary stored material.                                                                                                                                        |
| <b>LearningCandidate</b>            | An evaluated proposed module improvement; saving a <b>KnowledgeRecord</b> is not learning.                                                                                                                            |
| <b>ConversationRecord</b>           | Retained interaction material for continuity, traceability, or evaluation; it is not knowledge by default.                                                                                                            |

| Term               | Definition                                                                                                                                                                                                                             |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Read-first console | A local inspection surface that exposes status and <code>RuntimeJournal</code> trace without becoming a privileged mutation path.                                                                                                      |
| Recovery path      | The defined correction, rollback, deletion, compensation, or blocking behavior for an outcome or state transition.                                                                                                                     |
| External tool      | A non-canonical industry term for optional integrations outside the default <b>MADRE</b> internal-action path. External tools and services are higher-risk external actions requiring explicit policy, trace, and recovery boundaries. |